Differentiation

Chapter Overview

Gradient and first principles

Understand the derivative as a gradient function and derive simple derivatives from the limit definition.

Differentiation rules

Differentiate powers, sums and expressions rewritten as powers. Find second derivatives.

Using derivatives

Find tangents and normals, increasing and decreasing intervals, and stationary points. Sketch functions and their derivatives.

Modelling

Use rates of change and constraints to solve maximum and minimum problems.

Gradient Function

Solution 3

For a straight line, the gradient is constant:

$y = 3x + 2$

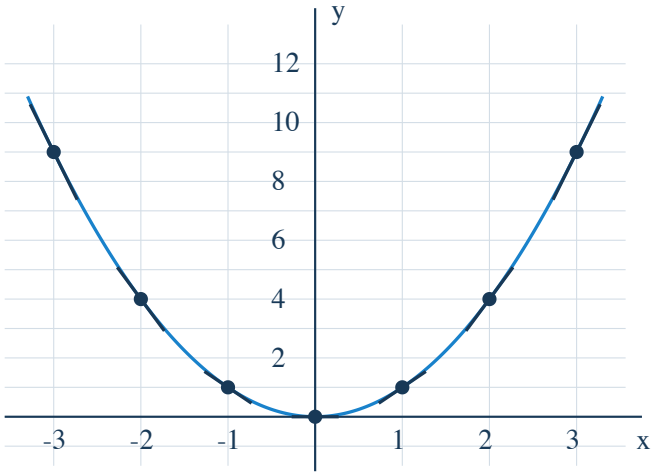


$m = 3$

For a curve, the gradient varies. We can no longer use a single constant for the gradient. We want an expression in x that gives the gradient at any value of x: the gradient function.

At GCSE, you found the gradient of a curve at a particular point by drawing a tangent.

$y = x^2$



x	-3	-2	-1	0	1	2	3
Gradient	1. -6	2. -4	3. -2	4. 0	5. 2	6. 4	7. 6

Use the relationship between x and the gradients to find the gradient function.

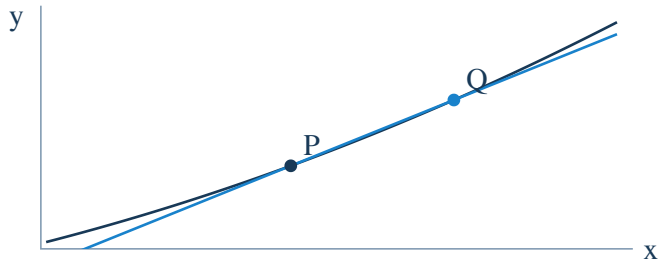
8. The gradient function is $2x$.

Finding the Gradient Function

Solution 4

Is there a method to find the gradient function without drawing lots of tangents and hoping to spot the rule?

To approximate the gradient on $y = x^2$ when $x = 5$, choose a second point slightly to the right. First use $x = 6$, then find the gradient between P and Q.



Coordinates of P

1. $(5, 25)$

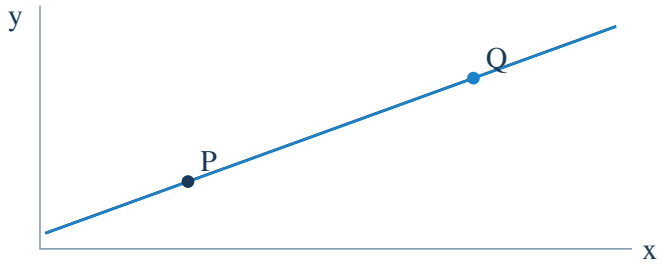
Coordinates of Q

2. $(6, 36)$

3. $m = \frac{\Delta y}{\Delta x} = \frac{36-25}{6-5} = \frac{11}{1} = 11.$

As the second point gets closer, the gradient between the points becomes a better approximation of the true gradient. Now use $x = 5.01$.

Magnified view near $x = 5$



P remains $(5, 25)$. Find the coordinates of Q.

4. $(5.01, 25.1001)$

5. $m = \frac{25.1001-25}{5.01-5} = \frac{0.1001}{0.01} = 10.01.$

The actual gradient at $x = 5$ is 10, so the second calculation gives a much closer approximation.

Finding the Gradient Function

Solution 5

For $y = x^2$, replace a numerical step by a small nonzero change h. Find the gradient at a general point.

Starting point

P

$$P = (x, x^2)$$

Nearby point

Q

$$Q = (x + h, (x + h)^2)$$

Form the difference quotient

$$\frac{(x + h)^2 - x^2}{h}$$

Expand and cancel, for h ≠ 0

$$\frac{2xh + h^2}{h} = 2x + h$$

Take the limit

$$f\prime(x) = \lim_{h \to 0}(2x + h) = 2x$$

lim means the limiting value as the input approaches a value. We simplify before taking the limit; we never divide by zero.

Finding the Gradient Function

Solution 6

The gradient function, or derivative

$$f\prime(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$$

This is differentiation from first principles. When the limit exists, it gives the gradient of the tangent at x.

Leibniz notation

$$y = x^2 \quad \Rightarrow \quad \frac{dy}{dx} = 2x$$

Lagrange notation

$$f(x) = x^2 \quad \Rightarrow \quad f\prime(x) = 2x$$

Changes and limits

A finite change is often written Δx or δx. Taking the limit of the difference quotient gives dy/dx.

$$\lim_{\delta x \to 0} \frac{f(x + \delta x) - f(x)}{\delta x}$$

Newton notation

A dot usually denotes differentiation with respect to time: $\dot{y} = dy/dt$. For $y = t^2$, $\dot{y} = 2t$.

Keep the variable of differentiation clear.

Example

Solution 7

A(4,16) lies on $y = x^2$, where the gradient is g.

(a) Show that $g = \lim_{h\to 0}(8 + h)$.

a

$$g = \lim_{h\to 0} \frac{(4+h)^2 - 16}{h}$$
$$= \lim_{h\to 0} \frac{8h + h^2}{h}$$
$$= \lim_{h\to 0}(8 + h)$$

(b) Deduce the value of g.

b

$g = 8$

As h tends to zero, 8 + h tends to 8.

Test Your Understanding

Solution 8

Prove from first principles that the derivative of x^4 is $4x^3$.

The relevant row of Pascal's triangle is 1, 4, 6, 4, 1.

Proof

$$f\prime(x) = \lim_{h\to 0} \frac{(x+h)^4 - x^4}{h}$$
$$= \lim_{h\to 0} \frac{4x^3h + 6x^2h^2 + 4xh^3 + h^4}{h}$$
$$= \lim_{h\to 0}(4x^3 + 6x^2h + 4xh^2 + h^3)$$
$$= 4x^3$$

After cancellation, every term containing a positive power of h tends to zero.

Exercise 12B – Pearson Pure Y1 page 261-262 Solution 9

5. Prove from first principles that the derivative of $6x$ is 6.

Use the definition

$$f\prime(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Proof

$$\begin{aligned} f\prime(x) &= \lim_{h\to 0} \frac{6(x+h) - 6x}{h} \\ &= \lim_{h\to 0} \frac{6h}{h} \\ &= \lim_{h\to 0} (6) \\ &= 6 \end{aligned}$$

Exercise 12B – Pearson Pure Y1 page 261-262 Solution 10

6. Prove from first principles that the derivative of $4x^2$ is $8x$.

Use the definition

$$f\prime(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Proof

$$\begin{aligned} f\prime(x) &= \lim_{h\to 0} \frac{4(x+h)^2 - 4x^2}{h} \\ &= \lim_{h\to 0} \frac{8xh + 4h^2}{h} \\ &= \lim_{h\to 0} (8x + 4h) \\ &= 8x \end{aligned}$$

Exercise 12B – Pearson Pure Y1 page 261-262 Solution 11

7. Prove from first principles that the derivative of ax^2 is $2ax$. Here a is constant.

Use the definition

$$f\prime(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Proof

$$\begin{aligned} f\prime(x) &= \lim_{h\to 0} \frac{a(x+h)^2 - ax^2}{h} \\ &= \lim_{h\to 0} \frac{2axh + ah^2}{h} \\ &= \lim_{h\to 0} (2ax + ah) \\ &= 2ax \end{aligned}$$

Exercise 12B – Pearson Pure Y1 page 261-262 Solution 12

Challenge: $f(x) = 1/x$, with $x \neq 0$.

(a) Use first principles to show that $f\prime(x) = \lim_{h\to 0} \frac{-1}{x^2+xh}$.

a

$$\begin{aligned} f\prime(x) &= \lim_{h\to 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\ &= \lim_{h\to 0} \frac{x - (x+h)}{xh(x+h)} \\ &= \lim_{h\to 0} \frac{-1}{x(x+h)} \end{aligned}$$

(b) Deduce that $f\prime(x) = -1/x^2$.

b

$$f\prime(x) = \lim_{h\to 0} \frac{-1}{x^2 + xh} = -\frac{1}{x^2}$$

Just for your interest... Solution 13

Why can we not immediately set $h = 0$ in $\lim_{h\to 0} \dfrac{(x+h)^2 - x^2}{h}$?

If we substitute $h = 0$ straight away:
$\lim_{h\to 0} \dfrac{(x+0)^2 - x^2}{0} = \lim_{h\to 0} \dfrac{0}{0}$.
Wait: this does not determine the limit.

$\frac{0}{0}$ is an indeterminate form.
Some expressions have limits that grow without bound, such as $\frac{1}{h}$ as $h \to 0^+$. But an indeterminate form alone does not determine a limit.
Zero divided by a non-zero number is zero; dividing by a number approaching zero can produce unbounded values. These observations do not resolve $\frac{0}{0}$.

Thankfully, there are techniques that turn a limit expression into one without an indeterminate form.
In this example, expand and simplify:
$\frac{(x+h)^2-x^2}{h} = 2x + h$, for $h \neq 0$.
Hence $\lim_{h\to 0}(2x + h) = 2x$; substituting here is valid. Other techniques are more advanced.
This is important when sketching harder functions.

Can you guess other indeterminate forms: expressions whose limiting values cannot be determined from the form alone?

1.
$\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$
0^0, 1^∞, ∞^0

Further source resource: DrFrostMaths, resource 163.

Differentiating x^n Solution 14

The power rule

$$\frac{d}{dx}(ax^n) = nax^{n-1}$$

Multiply by the power and reduce the power by one. Here a and n are constants; use a domain where the function is defined.

1. $y = x^5$

1
$\dfrac{dy}{dx} = 5x^4$

2. $f(x) = x^{1/2}$

2
$f\prime(x) = \frac{1}{2}x^{-1/2}$

3. $y = 2x^6$

3
$\dfrac{dy}{dx} = 12x^5$

4. $f(x) = x/x^4$

4
$f(x) = x^{-3}$, $f\prime(x) = -3x^{-4}$

5. $y = \sqrt{x^6}$, for $x \geq 0$.

5
$y = x^3$, $\dfrac{dy}{dx} = 3x^2$

The rule does not apply to 2^x: its exponent is variable, not constant.

Test Your Understanding

Solution 15

Differentiate each expression. In question 4, a is constant.

1. $y = x^7$

1
$\frac{dy}{dx} = 7x^6$

2. $y = 3x^{10}$

2
$\frac{dy}{dx} = 30x^9$

3. $f(x) = \frac{x^{1/2}}{x^2}$

3: rewrite
$f(x) = x^{-3/2}$

3: derivative
$f\prime(x) = -\frac{3}{2}x^{-5/2}$

4. $y = ax^a$

4
$\frac{dy}{dx} = a^2x^{a-1}$

5. $f(x) = \sqrt{49x^7}$

5: rewrite $f(x) = 7x^{7/2}$

5: derivative
$f\prime(x) = \frac{49}{2}x^{5/2}$

Differentiating Multiple Terms

Solution 16

Differentiate a sum or difference term by term.

$$\frac{d}{dx}[f(x) + g(x)] = f\prime(x) + g\prime(x)$$

Example: $y = x^2 + 4x + 3$.

$\frac{d}{dx}(x^2)$

$2x$

$\frac{d}{dx}(4x)$

4

$\frac{d}{dx}(3)$

3
0

Combine the derivatives

Result
$\frac{dy}{dx} = 2x + 4$

A linear term has a constant derivative. A constant term has derivative zero because its graph is horizontal.

Quickfire Questions

Solution 17

Differentiate each function. Here a and p are constants.

1. $y = 2x^2 - 3x$

1
$$\frac{dy}{dx} = 4x - 3$$

2. $y = 4 - 9x^3$

2
$$\frac{dy}{dx} = -27x^2$$

3. $y = 5x + 1$

3
$$\frac{dy}{dx} = 5$$

4. $y = ax$

4
$$\frac{dy}{dx} = a$$

5. $y = 6x - 3 + px^2$

5
$$\frac{dy}{dx} = 6 + 2px$$

Harder Example

Solution 18

Let $f(x) = 4x^2 - 8x + 3$.

(a) Find the gradient at $(\frac{1}{2}, 0)$.

a
$$f\prime(x) = 8x - 8$$
$$f\prime(\tfrac{1}{2}) = -4$$

(b) Find the point where the gradient is 8.

b
$$8x - 8 = 8 \Rightarrow x = 2$$
$$f(2) = 3 \Rightarrow (2, 3)$$

(c) Find the gradients where the curve meets $y = 4x - 5$.

c
$$4x^2 - 8x + 3 = 4x - 5$$
$$4(x - 1)(x - 2) = 0$$
$$f\prime(1) = 0, \quad f\prime(2) = 8$$

A known x gives the gradient. A known gradient gives an equation for x; substitute into the original function to find y.

Test Your Understanding — Solution 19

Let $f(x) = x^2 - 4x + 2$.

(a) Find the gradient at $(1, -1)$.

a

$f\prime(x) = 2x - 4$

$f\prime(1) = -2$

(b) Find the point where the gradient is 5.

b

$2x - 4 = 5 \Rightarrow x = \frac{9}{2}$

$f(\frac{9}{2}) = \frac{17}{4}$

The point is $(\frac{9}{2}, \frac{17}{4})$.

(c) Find the gradients where the curve meets $y = 2 - x$.

c

$x^2 - 4x + 2 = 2 - x$

$x(x - 3) = 0$

$f\prime(0) = -4, \quad f\prime(3) = 2$

A known x gives the gradient. A known gradient gives an equation for x; substitute into the original function to find y.

Exercise 12D – Pearson Pure Y1 page 265-266 — Solution 20

5. Find the gradients of $y = x^2 - 5x + 10$ where the curve meets $y = 4$.

Find the intersections

$x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0$

Differentiate

$\frac{dy}{dx} = 2x - 5$

Evaluate at each intersection

$x = 2: \ m = -1; \qquad x = 3: \ m = 1$

Exercise 12D – Pearson Pure Y1 page 265-266 Solution 21

6. Find the gradients of $y = 2x^2$ where the curve meets $y = x + 3$.

Find the intersections

$2x^2 - x - 3 = 0 \Rightarrow (2x - 3)(x + 1) = 0$

Differentiate

$\frac{dy}{dx} = 4x$

Evaluate at each intersection

$x = -1 : \ m = -4; \qquad x = \frac{3}{2} : \ m = 6$

Exercise 12D – Pearson Pure Y1 page 265-266 Solution 22

7. $f(x) = x^2 - 2x - 8$.

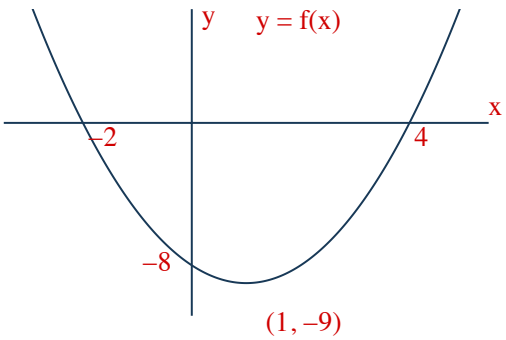
a. Sketch $y = f(x)$.

b. On the same axes, sketch $y = f\prime(x)$.

c. Explain why the turning point of $y = f(x)$ has the same x-coordinate as the point where $y = f\prime(x)$ crosses the x-axis.

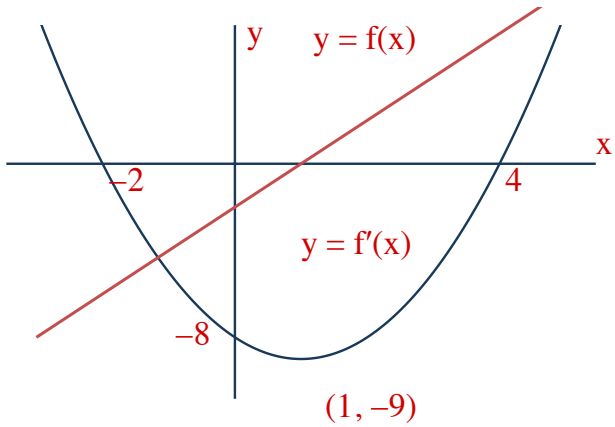
a.

a. As the coefficient of x^2 is positive, the parabola opens upwards.
$f(0) = -8$. When $y = 0$, $(x + 2)(x - 4) = 0$, so $x = -2, 4$.
Completing the square:
$x^2 - 2x - 8 = (x - 1)^2 - 1 - 8 = (x - 1)^2 - 9$.
The minimum is $(1, -9)$.



b.

b.
$f\prime(x) = 2x - 2 + 0 = 2x - 2$.
The straight line crosses the axes at $(1, 0)$ and $(0, -2)$.



c. c. At a turning point, the gradient is zero: $f\prime(x) = 0$. Here $2x - 2 = 0$, so both points have $x = 1$.

Differentiating Harder Expressions

Solution 23

Rewrite as a sum of powers of x before differentiating.

1. Turn roots into powers

$y = \sqrt{x}$

1: rewrite $y = x^{1/2}$

1: derivative $\frac{dy}{dx} = \frac{1}{2}x^{-1/2}$

2. Turn roots into powers

$y = 1/\sqrt[3]{x}$

2: rewrite $y = x^{-1/3}$

2: derivative $\frac{dy}{dx} = -\frac{1}{3}x^{-4/3}$

3. Split fractions

$y = \frac{x^2+3}{\sqrt{x}}$

3: rewrite $y = x^{3/2} + 3x^{-1/2}$

3: derivative $\frac{dy}{dx} = \frac{3}{2}x^{1/2} - \frac{3}{2}x^{-3/2}$

4. Expand brackets

$y = x^2(x-3)$

4: rewrite $y = x^3 - 3x^2$

4: derivative $\frac{dy}{dx} = 3x^2 - 6x$

5. Keep constant factors

$y = \frac{1}{3x}$

5: rewrite $y = \frac{1}{3}x^{-1}$

5: derivative $\frac{dy}{dx} = -\frac{1}{3}x^{-2}$

Test Your Understanding

Solution 24

Differentiate each expression.

1. $y = \frac{1}{\sqrt{x}}$

1: rewrite $y = x^{-1/2}$

1: derivative $\frac{dy}{dx} = -\frac{1}{2}x^{-3/2}$

2. $y = \frac{2+x^3}{x^2}$

2: rewrite $y = 2x^{-2} + x$

2: derivative $\frac{dy}{dx} = -4x^{-3} + 1$

3. $y = \frac{1+2x}{3x\sqrt{x}}$

3: rewrite $y = \frac{1}{3}x^{-3/2} + \frac{2}{3}x^{-1/2}$

3: derivative $\frac{dy}{dx} = -\frac{1}{2}x^{-5/2} - \frac{1}{3}x^{-3/2}$

Exercise 12E – Pearson Pure Y1 page 267-268 Solution 25

6. $f(x) = \frac{12}{p\sqrt{x}} + x$, where p is a nonzero real constant and x > 0. Given $f\prime(2) = 3$, find p in the form $a\sqrt{2}$, where a is rational.

Rewrite and differentiate

$$f(x) = \frac{12}{p}x^{-1/2} + x$$
$$f\prime(x) = -\frac{6}{p}x^{-3/2} + 1$$

Use the given gradient

$$-\frac{6}{2p\sqrt{2}} + 1 = 3$$
$$p = -\frac{3}{2\sqrt{2}} = -\frac{3}{4}\sqrt{2}$$

Exercise 12E – Pearson Pure Y1 page 267-268 Solution 26

7. $f(x) = (2 - x)^9$.

(a) Find the first three terms in ascending powers of x.

a

$$(2 - x)^9 = 2^9 + 9(2^8)(-x) + 36(2^7)x^2 + \cdots$$
$$= 512 - 2304x + 4608x^2 + \cdots$$

(b) For small x, ignore terms in x^2 and higher in the derivative. Show that $f\prime(x) \approx 9216x - 2304$.

b

$$f\prime(x) = -2304 + 9216x + \text{terms in } x^2 \text{ and higher}$$
$$f\prime(x) \approx 9216x - 2304$$

Extension — Solution 27

[MAT 2013 1E]

The expression $\frac{d^2}{dx^2}\left[(2x-1)^4(1-x)^5\right] + \frac{d}{dx}\left[(2x+1)^4(3x^2-2)^2\right]$ is a polynomial of degree:

9

8

7

less than 7

The second derivative means differentiate twice. It comes later in this chapter.

Full expansion is not needed. The leading term in the first polynomial is $-16x^9$, whose second derivative is $-1152x^7$.

The leading term in the second polynomial is $144x^8$, whose derivative is $1152x^7$.

These cancel, leaving degree at most 6: answer D.

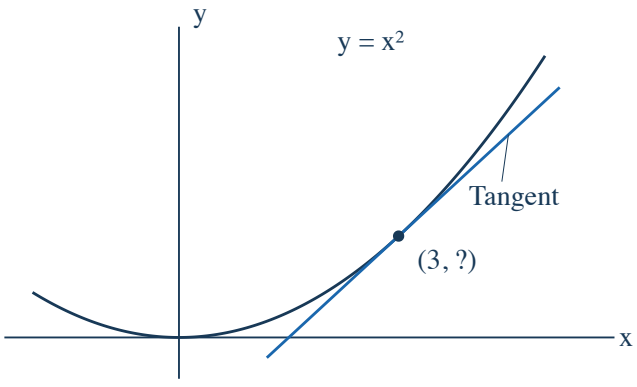
Finding equations of tangents — Solution 28

Find the equation of the tangent to the curve $y = x^2$ when $x = 3$.

We want to use $y - y_1 = m(x - x_1)$ for the tangent, because it is a straight line.

Therefore we need:

- A point (x_1, y_1).
- The gradient m.



Gradient function:

1. $\frac{dy}{dx} = 2x$

Gradient when $x = 3$:

2. $m = 2(3) = 6$

y-value when $x = 3$:

3. $y = 3^2 = 9$

So the equation of the tangent is:

4. $y - 9 = 6(x - 3)$.

Equivalently, $y = 6x - 9$.

Finding equations of normals

Solution 29

Find the equation of the normal to $y = x^2$ at x = 3.

The normal is perpendicular to the tangent at the point of contact.

Point and tangent gradient

$P = (3, 9), \qquad m_T = 2(3) = 6$

Normal gradient

$$m_N = -\frac{1}{m_T} = -\frac{1}{6}$$

The tangent is $y - 9 = 6(x - 3)$.

Equation of the normal

$$y - 9 = -\frac{1}{6}(x - 3)$$

Check whether the question asks for a tangent or a normal. Their gradients are negative reciprocals when both are finite and nonzero.

Test Your Understanding

Solution 30

Find the normal to $y = x + 3\sqrt{x}$ at x = 9.

Point on the curve

$y = 9 + 3\sqrt{9} = 18$

Tangent and normal gradients

$$\frac{dy}{dx} = 1 + \frac{3}{2}x^{-1/2}$$

$$m_T = \frac{3}{2}, \qquad m_N = -\frac{2}{3}$$

Normal equation

$$y - 18 = -\frac{2}{3}(x - 9)$$

Exercise 12F – Pearson Pure Y1 page 269-270 Solution 31

3. Find where the tangent to $y = x^2 + 1$ at $(2,5)$ meets the normal to the curve at $(1,2)$.

Tangent at $(2,5)$

$$\frac{dy}{dx} = 2x, \quad m_T = 4$$
$$y - 5 = 4(x - 2) \Rightarrow y = 4x - 3$$

Normal at $(1,2)$

$$m_N = -\frac{1}{2}$$
$$y - 2 = -\frac{1}{2}(x - 1) \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$$

Solve simultaneously

3

$$4x - 3 = -\frac{1}{2}x + \frac{5}{2} \Rightarrow x = \frac{11}{9}$$
$$y = \frac{17}{9} \Rightarrow \left(\frac{11}{9}, \frac{17}{9}\right)$$

Exercise 12F – Pearson Pure Y1 page 269-270 Solution 32

4. Find the normals to $y = x + x^3$ at $(0,0)$ and $(1,2)$, and find their point of intersection.

Normal at $(0,0)$

$$\frac{dy}{dx} = 1 + 3x^2, \quad m_T = 1, \quad m_N = -1$$
$$y = -x$$

Normal at $(1,2)$

$$m_T = 4, \quad m_N = -\frac{1}{4}$$
$$y - 2 = -\frac{1}{4}(x - 1) \Rightarrow 4y + x - 9 = 0$$

Intersection

$$4(-x) + x - 9 = 0 \Rightarrow x = -3, \quad y = 3$$
$$(-3, 3)$$

Exercise 12F – Pearson Pure Y1 page 269-270 Solution 33

5. For $f(x) = 12 - 4x + 2x^2$, find the tangent and normal at x = −1.

Point and gradient

$f(-1) = 18, \quad f\prime(x) = 4x - 4$

$f\prime(-1) = -8$

Tangent

$y - 18 = -8(x + 1) \Rightarrow y = -8x + 10$

Normal

$y - 18 = \frac{1}{8}(x + 1) \Rightarrow 8y - x - 145 = 0$

Exercise 12F – Pearson Pure Y1 page 269-270 Solution 34

6. P has x-coordinate $\frac{1}{2}$ on $y = 2x^2$. The normal at P meets the curve again at Q. Find Q. (6 marks)

Normal at P

$P = (\frac{1}{2}, \frac{1}{2}), \quad \frac{dy}{dx} = 4x$

$m_T = 2, \quad m_N = -\frac{1}{2}$

$y = -\frac{1}{2}x + \frac{3}{4}$

Second intersection

$2x^2 = -\frac{1}{2}x + \frac{3}{4}$

$8x^2 + 2x - 3 = (4x + 3)(2x - 1) = 0$

$x = -\frac{3}{4} \Rightarrow y = \frac{9}{8}$

$Q = \left(-\frac{3}{4}, \frac{9}{8}\right)$

Extension

Solution 35

1. [STEP I 2005 Q2] On $y^2 = 4x$, let $P = (p^2, 2p)$ and $Q = (q^2, 2q)$, where p and q are nonzero and distinct.

(a) Show that their tangents meet at $R = (pq, p+q)$.

(b) Their normals meet at S. If (1,0) lies on PQ, show that $S = (p^2+q^2+1, p+q)$ and that PSQR is a rectangle.

Worked solution on the next slide.

2. [STEP I 2012 Q4] On $xy = \frac{1}{2}$, let $P = (p, \frac{1}{2p})$, $Q = (q, \frac{1}{2q})$, with distinct positive p and q.

(a) Show that the tangents meet at $T = (\frac{2pq}{p+q}, \frac{1}{p+q})$.

(b) When $pq = \frac{1}{2}$, find the intersection N of the normals. Show that T and N lie on y = x and that their distances from O have constant product.

2(a)

$$y = -\frac{x}{2p^2} + \frac{1}{p}, \qquad y = -\frac{x}{2q^2} + \frac{1}{q}$$

$$T = \left(\frac{2pq}{p+q}, \frac{1}{p+q}\right)$$

2(b)

$$y = 2p^2x + \frac{1}{2p} - 2p^3$$

$$N = (p+q, p+q), \quad T = (\tfrac{1}{p+q}, \tfrac{1}{p+q})$$

$$ON \cdot OT = \sqrt{2}(p+q)\frac{\sqrt{2}}{p+q} = 2$$

Useful identity

$$p^3 - q^3 = (p-q)(p^2+pq+q^2)$$

Solutions to Extension Question 1

Solution 36

For $y^2 = 4x$, implicit differentiation gives $2y\,dy/dx = 4$, so $dy/dx = 2/y$. This works on either branch when y ≠ 0.

Tangents at P and Q

$$py = x + p^2, \qquad qy = x + q^2$$

$$(q-p)y = q^2 - p^2 \Rightarrow y = p+q$$

$$x = pq \Rightarrow R = (pq, p+q)$$

Normals at P and Q

$$y + px = 2p + p^3, \qquad y + qx = 2q + q^3$$

$$x = p^2 + pq + q^2 + 2, \quad y = -pq(p+q)$$

Use the chord condition

$$PQ: \quad (p+q)y = 2x + 2pq$$

Substitute (1,0):

$$0 = 2 + 2pq \Rightarrow pq = -1$$

$$S = (p^2+q^2+1, p+q)$$

Show that PSQR is a rectangle

Tangent–normal pairs give right angles at P and Q. The tangent gradients have product

$$\frac{1}{p}\frac{1}{q} = \frac{1}{pq} = -1.$$

The normal gradients have product

$$(-p)(-q) = pq = -1.$$

So the angles at R and S are also right angles. PSQR is a rectangle.

Increasing and Decreasing Functions

Solution 37

Increasing and decreasing

A function is increasing on an interval when larger inputs give values that are at least as large. For a differentiable function, $f\prime(x) \geq 0$ throughout an interval ensures this.

Likewise, $f\prime(x) \leq 0$ throughout an interval ensures the function is decreasing.

Strictly increasing

If $f\prime(x) > 0$ throughout an interval, the function is strictly increasing. This is sufficient, not necessary: x^3 is strictly increasing although its derivative is zero at $x = 0$.

A curve has a local maximum at $(2,3)$ and a local minimum at $(4,-1)$. Its gradient is positive outside these turning points and negative between them. State its increasing and decreasing intervals.

Increasing: left

$x \leq 2$

Decreasing

$2 \leq x \leq 4$

Increasing: right

$x \geq 4$

Interval notation: $[2,4] = \{x : 2 \leq x \leq 4\}$. Square brackets include the endpoints.

Examples

Solution 38

1. Show that $f(x) = x^3 + 6x^2 + 21x + 2$ is increasing for every real x.

2. Find the interval where $g(x) = x^3 + 3x^2 - 9x$ is decreasing.

1

$f\prime(x) = 3x^2 + 12x + 21$

$= 3(x+2)^2 + 9 > 0$

A square is nonnegative, so the derivative is positive for every real x. Hence f is strictly increasing.

2

$g\prime(x) = 3x^2 + 6x - 9$

$= 3(x+3)(x-1)$

Solve $g\prime(x) \leq 0$:

$x \in [-3, 1]$

Test Your Understanding — Solution 39

1. Show that $f(x) = x^3 + 16x - 2$ is increasing for every real x.

1

$f\prime(x) = 3x^2 + 16$

$= 3x^2 + 16 > 0$

A square is nonnegative, so the derivative is positive for every real x. Hence f is strictly increasing.

2. Find the interval where $g(x) = x^3 + 6x^2 - 135x$ is decreasing.

2

$g\prime(x) = 3x^2 + 12x - 135$

$= 3(x + 9)(x - 5)$

Solve $g\prime(x) \leq 0$:

$x \in [-9, 5]$

Exercise 12G – Pearson Pure Y1 page 271 — Solution 40

3. Show that $f(x) = 4 - x(2x^2 + 3)$ is decreasing for all real x.

Expand and differentiate

$f(x) = 4 - 2x^3 - 3x$

$f\prime(x) = -6x^2 - 3$

Determine the sign

$x^2 \geq 0 \Rightarrow -6x^2 - 3 \leq -3 < 0$

The derivative is negative for all real x, so f is strictly decreasing.

Exercise 12G – Pearson Pure Y1 page 271 Solution 41

4. $f(x) = x^2 + px$ is increasing on $[-1,1]$.

(a) Find one possible value of p.

(b) Is this the only possible value? Justify your answer.

a

$f\prime(x) = 2x + p$

The smallest derivative on this interval occurs at x $= -1$.

$p - 2 \geq 0 \Rightarrow p \geq 2$.

One possible value is p $= 3$.

b

No. Every $p \geq 2$ gives $2x + p \geq 0$ for all $x \in [-1, 1]$. Hence every such p works.

Second Order Derivatives Solution 42

The second derivative is the derivative of the first derivative. Keep the variable of differentiation consistent.

Leibniz notation	Lagrange notation	Newton notation: time
$y = x^4$	$f(x) = x^4$	$y = t^4$
$\frac{dy}{dx} = 4x^3$	$f\prime(x) = 4x^3$	$\dot{y} = 4t^3$
$\frac{d^2y}{dx^2} = 12x^2$	$f\prime\prime(x) = 12x^2$	$\ddot{y} = 12t^2$

The second derivative describes how the gradient changes. We will use its sign to help classify stationary points.

Just for your interest... Solution 43

Why is the second derivative written this way?

For $y = x^3 + 1$, the symbol d/dx is an operator: it means "differentiate with respect to x".

$$\frac{d}{dx}(x^3 + 1) = 3x^2$$

Apply the operator twice

$$\frac{d}{dx}\left(\frac{d}{dx}(y)\right) = \frac{d^2y}{dx^2}$$

Example

$$\frac{d}{dx}(3x^2) = 6x$$

$$\frac{d^2y}{dx^2} = 6x$$

The superscripts indicate two applications of differentiation with respect to x. Do not treat the notation as an ordinary fraction to be squared.

Examples Solution 44

1. If $y = 3x^5 + \frac{4}{x^2}$, find d^2y/dx^2.

1: first derivative

$$y = 3x^5 + 4x^{-2}$$

$$\frac{dy}{dx} = 15x^4 - 8x^{-3}$$

1: second derivative

$$\frac{d^2y}{dx^2} = 60x^3 + 24x^{-4}$$

2. If $f(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}$, find $f''(x)$.

2: first derivative

$$f(x) = 3x^{1/2} + \tfrac{1}{2}x^{-1/2}$$

$$f'(x) = \tfrac{3}{2}x^{-1/2} - \tfrac{1}{4}x^{-3/2}$$

2: second derivative

$$f''(x) = -\tfrac{3}{4}x^{-3/2} + \tfrac{3}{8}x^{-5/2}$$

Test Your Understanding

Solution 45

If $y = 5x^3 - \frac{x}{3\sqrt{x}}$, find d^2y/dx^2.

Rewrite as powers

$y = 5x^3 - \frac{1}{3}x^{1/2}$

Differentiate once

$\frac{dy}{dx} = 15x^2 - \frac{1}{6}x^{-1/2}$

Differentiate again

$\frac{d^2y}{dx^2} = 30x + \frac{1}{12}x^{-3/2}$

Exercise 12H – Pearson Pure Y1 page 272

Solution 46

3. Given $y = (2x - 3)^3$, find x when $d^2y/dx^2 = 0$.

3

$y = 8x^3 - 36x^2 + 54x - 27$

$\frac{dy}{dx} = 24x^2 - 72x + 54$

3

$\frac{d^2y}{dx^2} = 48x - 72$

$48x - 72 = 0 \Rightarrow x = \frac{3}{2}$

Exercise 12H – Pearson Pure Y1 page 272 Solution 47

4. For $y = px^3 - 3px^2 + x^2 - 4$, the second derivative at x = 2 is -1. Find p.

4

$$\frac{dy}{dx} = 3px^2 - 6px + 2x$$

$$\frac{d^2y}{dx^2} = 6px - 6p + 2$$

4

$$12p - 6p + 2 = -1$$

$$6p = -3 \Rightarrow p = -\frac{1}{2}$$

Stationary/Turning Points Solution 48

A stationary point is a point where the derivative is zero. It may be a local maximum, a local minimum or a stationary point of inflection.

Find the stationary points of $y = x^3 + 6x^2 - 135x$.

Solve for the x-coordinates

$$\frac{dy}{dx} = 3x^2 + 12x - 135$$

$$3(x + 9)(x - 5) = 0 \Rightarrow x = -9, 5$$

Find the y-coordinates

$$x = -9 \Rightarrow y = 972$$

$$x = 5 \Rightarrow y = -400$$

$$(-9, 972), \quad (5, -400)$$

Use the original function to find y, not the derivative. We will next classify the two points.

More Examples

Solution 49

1. Find the least value of
$f(x) = x^2 - 4x + 9$.

1. Method 1

Method 1: Differentiation

$f\prime(x) = 2x - 4 = 0.$

$x = 2.$

$f(2) = 2^2 - 4(2) + 9 = 5.$

Therefore the minimum value is 5.

1. Method 2

Method 2: Completing the square

$f(x) = (x-2)^2 + 5.$

Since $(x-2)^2 \geq 0$, the minimum value of f(x) is 5. This occurs at $x = 2$.

Note: Method 2 applies to quadratic functions. For other functions, use differentiation.

2. Find the turning point of
$y = \sqrt{x} - x$.

2.

$y = x^{1/2} - x.$

$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} - 1 = 0.$

$\frac{1}{2}x^{-1/2} = 1.$

$x^{-1/2} = 2 \quad \Rightarrow \quad \frac{1}{\sqrt{x}} = 2.$

$\sqrt{x} = \frac{1}{2} \quad \Rightarrow \quad x = \frac{1}{4}.$

When $x = \frac{1}{4}$,

$y = \sqrt{\frac{1}{4}} - \frac{1}{4} = \frac{1}{4}.$

The turning point is $\left(\frac{1}{4}, \frac{1}{4}\right)$.

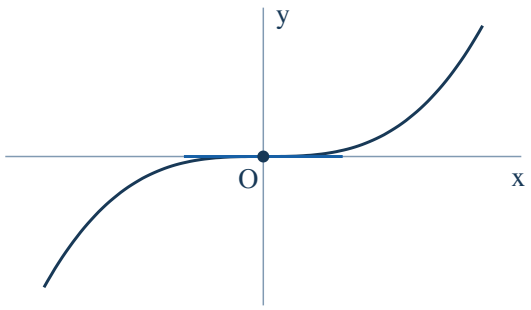
Points of Inflection

Solution 50

A third type of stationary point is a stationary point of inflection.

A point of inflection is where a curve changes concavity: it curves one way before the point, then the other way after it.

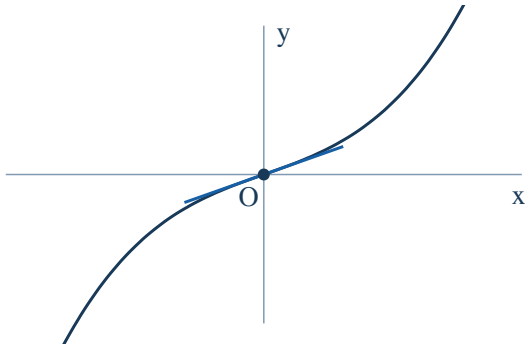
Stationary point of inflection



At the point, $f\prime(x) = 0$.
The tangent is horizontal; the concavity changes.

The terms convex and concave are also used in optics. Terminology varies; the key feature here is the change in concavity.

Non-stationary point of inflection



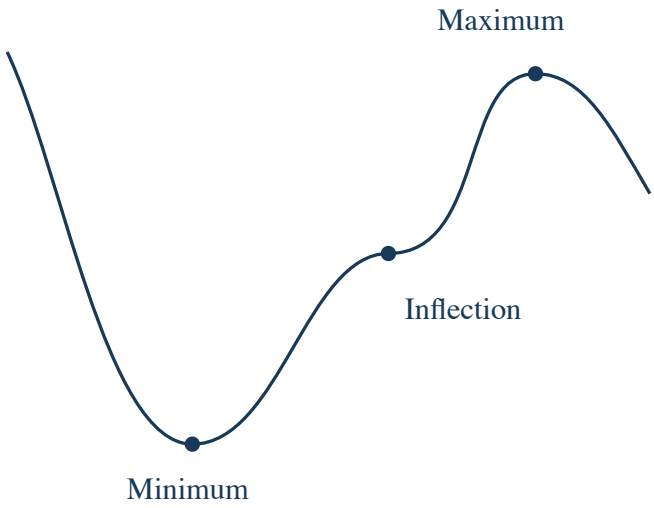
Here $f\prime(x) \neq 0$.
Not every point of inflection is stationary.

A stationary point of inflection is sometimes called a saddle point.

How do we tell what type of stationary point?

Solution 51

Method 1: Compare the gradient just before and just after the stationary point.



At a minimum or maximum the gradient changes sign. At a stationary point of inflection it is zero at the point without a change of sign.

Local minimum - gradient

Before point	At point	After point
Negative	Zero	Positive

Local maximum - gradient

Before point	At point	After point
Positive	Zero	Negative

Stationary point of inflection - gradient

Before point	At point	After point
7. Positive	8. Zero	9. Positive

How do we tell what type of stationary point?

Solution 52

Find and classify the stationary point of $y = x^4 - 32x$ using the gradient on either side.

Locate the point

$$\frac{dy}{dx} = 4x^3 - 32 = 0 \Rightarrow x = 2$$
$$y = 2^4 - 32(2) = -48 \Rightarrow (2, -48)$$

Just before: x = 1.9

$4(1.9)^3 - 32 = -4.564 < 0$

At the point: x = 2

$4(2)^3 - 32 = 0$

Just after: x = 2.1

$4(2.1)^3 - 32 = 5.044 > 0$

Conclusion

The gradient changes from negative to positive, so $(2,-48)$ is a local minimum.

Method 2: Using the second derivative

Solution 53

The second derivative is the rate of change of the gradient.

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Near a local maximum

The gradient falls from positive to zero to negative. When the second derivative at the stationary point is negative, the point is a local maximum.

Near a local minimum

The gradient rises from negative to zero to positive. When the second derivative at the stationary point is positive, the point is a local minimum.

First locate a stationary point by solving $dy/dx = 0$. Then evaluate the second derivative at that x-coordinate.

Method 2: Using the second derivative

Solution 54

The second derivative test at a stationary point

$f''(x) > 0$

Local minimum: the gradient is increasing.

$f''(x) < 0$

Local maximum: the gradient is decreasing.

$f''(x) = 0$

Inconclusive. Check the gradient on either side or use another argument.

Example: $y = x^4 - 32x$ has a stationary point at $(2, -48)$.

Example

$$\frac{d^2y}{dx^2} = 12x^2$$

$$12(2)^2 = 48 > 0$$

Therefore $(2, -48)$ is a local minimum.

A zero second derivative does not automatically mean a point of inflection.

Test Your Understanding — Solution 55

The curve $y = x^2 - 32\sqrt{x} + 20$, $x > 0$, has a stationary point P. Use calculus:

a. Find the coordinates of P. (6 marks)

b. Determine the nature of P. (3 marks)

a.

a.

$\frac{dy}{dx} = 2x - 16x^{-1/2}$.

At a stationary point, $2x - 16x^{-1/2} = 0$.

Since $x > 0$, multiply by $\sqrt{x}$: $2x^{3/2} = 16$, so $x^{3/2} = 8$ and $x = 4$.

$y = 4^2 - 32\sqrt{4} + 20 = -28$. Therefore $P = (4, -28)$.

b.

b.

$\frac{d^2y}{dx^2} = 2 + 8x^{-3/2}$.

At $x = 4$, $\frac{d^2y}{dx^2} = 2 + 8(4^{-3/2}) = 3 > 0$. Therefore P is a minimum.

Sketching Graphs — Solution 56

In Chapter 4, we used intercepts and behaviour as $x \to \infty$ and $x \to -\infty$ to sketch graphs. Now we can also find stationary and turning points.

[Textbook] By first finding the stationary points, sketch $y = \frac{1}{x} + 27x^3$.

1.

$y = x^{-1} + 27x^3$.

$\frac{dy}{dx} = -x^{-2} + 81x^2 = 0$.

$-\frac{1}{x^2} + 81x^2 = 0 \Rightarrow 81x^2 = \frac{1}{x^2}$.

$81x^4 = 1 \Rightarrow x = \frac{1}{3}$ or $x = -\frac{1}{3}$.

The stationary points are $\left(\frac{1}{3}, 4\right)$ and $\left(-\frac{1}{3}, -4\right)$.

2. As $x \to \infty$, $y \to \infty$.

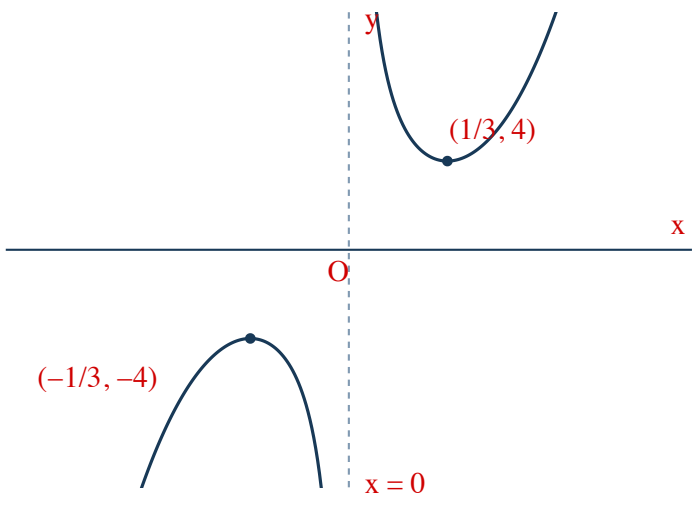
As $x \to -\infty$, $y \to -\infty$.

3.

The function is undefined at $x = 0$, because of $\frac{1}{x}$. Thus $x = 0$ is a vertical asymptote.

As $x \to 0^+$, $y \to \infty$; as $x \to 0^-$, $y \to -\infty$.

4.



Exercise 12I – Pearson Pure Y1 page 276 Solution 57

5. $f(x) = x^3 - 3x^2 + 3x$.

(a) Find the stationary point.

(b) Determine its nature.

a

$f\prime(x) = 3x^2 - 6x + 3 = 3(x-1)^2$

$f\prime(x) = 0 \Rightarrow x = 1, \quad f(1) = 1$

b

$f\prime(x) > 0$ on both sides of $x = 1$

$f\prime\prime(x) = 6(x-1)$

The gradient stays positive on both sides, while concavity changes. (1,1) is a stationary point of inflection.

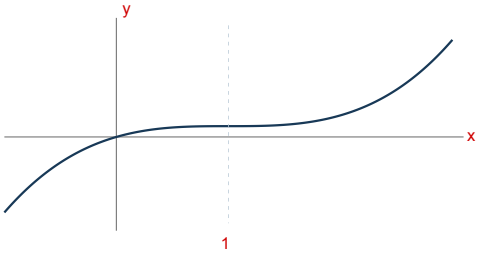
(c) Sketch the graph.

c

$f(x) = (x-1)^3 + 1$

Translate the graph of $y = x^3$ one unit right and one unit up. It passes through (0,0) and has a horizontal tangent at (1,1).

Sketch



Exercise 12I – Pearson Pure Y1 page 276 Solution 58

6. $f(x) = 27 - 2x^4$.

(a) Find and classify the stationary point.

(b) State the range of f.

a

$f\prime(x) = -8x^3 = 0 \Rightarrow x = 0$

$f(0) = 27$

The derivative is positive for $x < 0$ and negative for $x > 0$. Therefore (0,27) is a maximum.

b

$x^4 \geq 0 \Rightarrow 27 - 2x^4 \leq 27$

Range: $(-\infty, 27]$

Here $f\prime\prime(0) = 0$, so the second derivative test alone is inconclusive. The gradient sign change establishes the maximum.

Exercise 12I – Pearson Pure Y1 page 276 — Solution 59

7. $f(x) = x^4 + 3x^3 - 5x^2 - 3x + 1$.

a. Find the coordinates of the stationary points and determine the nature of each.

b. Sketch $y = f(x)$.

a.

$f\prime(x) = 4x^3 + 9x^2 - 10x - 3$.

Put $4x^3 + 9x^2 - 10x - 3 = 0$.

$f\prime(1) = 4 + 9 - 10 - 3 = 0$.

By the factor theorem, divide by x - 1:

	$4x^2$	$+13x$	$+3$	
$x - 1$	$4x^3$	$+9x^2$	$-10x$	-3
	$\underline{4x^3}$	$\underline{-4x^2}$		
		$13x^2$	$-10x$	
		$\underline{13x^2}$	$\underline{-13x}$	
			$3x$	-3
			$\underline{3x}$	$\underline{-3}$
				0

$(x - 1)(4x^2 + 13x + 3) = 0$.

$(x - 1)(4x + 1)(x + 3) = 0$.

$x = 1, -\frac{1}{4}$ or -3.

$f(1) = 1 + 3 - 5 - 3 + 1 = -3$.

$f(-\frac{1}{4}) = (-\frac{1}{4})^4 + 3(-\frac{1}{4})^3 - 5(-\frac{1}{4})^2 - 3(-\frac{1}{4}) + 1 = \frac{357}{256}$.

$f(-3) = (-3)^4 + 3(-3)^3 - 5(-3)^2 - 3(-3) + 1 = -35$.

$f\prime\prime(x) = 12x^2 + 18x - 10$.

$f\prime\prime(1) = 12 + 18 - 10 = 20 > 0$.

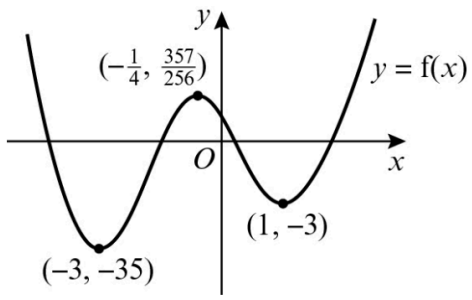
Minimum at $(1, -3)$.

$f\prime\prime(-3) = 12(-3)^2 + 18(-3) - 10 = 44 > 0$: minimum at $(-3, -35)$.

$f\prime\prime(-\frac{1}{4}) = 12(-\frac{1}{4})^2 + 18(-\frac{1}{4}) - 10 = -\frac{55}{4} < 0$.

Maximum at $(-\frac{1}{4}, \frac{357}{256})$.

b.



Extension — Solution 60

1. [MAT 2014 1C] The cubic
$y = kx^3 - (k+1)x^2 + (2-k)x - k$
has a turning point that is a minimum at $x = 1$ precisely for:
A) $k > 0$ B) $0 < k < 1$ C) $k > \frac{1}{2}$
D) $k < 3$ E) all values of k.

2. [MAT 2004 1B] Find the smallest value of
$f(x) = 2x^3 - 9x^2 + 12x + 3$
in the range $0 \le x \le 2$.

3. [MAT 2001 1E] The maximum gradient of
$y = x^4 - 4x^3 + 4x^2 + 2$
in $0 \le x \le 2\frac{1}{5}$ occurs when:
A) $x = 0$ B) $x = 1 - \frac{1}{\sqrt{3}}$
C) $x = 1 + \frac{1}{\sqrt{3}}$ D) $x = 2\frac{1}{5}$.

4. [STEP I 2007 Q8] A curve is given by
$y = ax^3 - 6ax^2 + (12a + 12)x - (8a + 16)$,
where a is a real number. Show that it touches $y = x^3$ at $(2, 8)$.
Determine the coordinates of any other point of intersection of the curves.
Sketch both curves on the same axes when:
(i) $a = 2$;
(ii) $a = 1$;
(iii) $a = -2$.

Hint: When two curves touch, their y-values must match. What else must also match?

Worked solutions follow on slides 61–63.

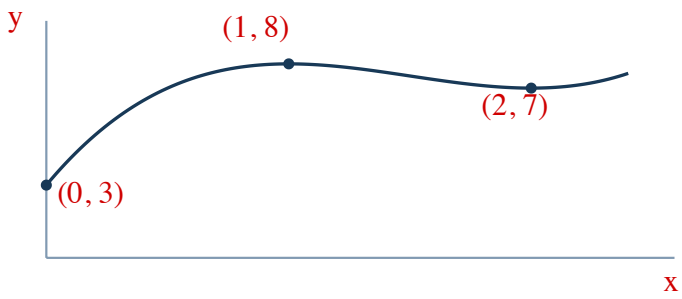
Solutions to Extension Questions

1. [MAT 2014 1C] The cubic
$y = kx^3 - (k+1)x^2 + (2-k)x - k$
has a turning point that is a minimum at $x = 1$ precisely for:
A) $k > 0$ B) $0 < k < 1$ C) $k > \frac{1}{2}$
D) $k < 3$ E) all values of k.

1.
$\frac{dy}{dx} = 3kx^2 - 2(k+1)x + (2-k)$.
At $x = 1$,
$\frac{dy}{dx} = 3k - 2k - 2 + 2 - k = 0$.
Thus $x = 1$ is stationary for every k. We require a minimum.
$\frac{d^2y}{dx^2} = 6kx - 2(k+1)$.
At $x = 1$, $\frac{d^2y}{dx^2} = 4k - 2$.
For a minimum, $4k - 2 > 0$, so $k > \frac{1}{2}$.
At $k = \frac{1}{2}$, the stationary point is an inflection, so it is excluded. Answer: C.

2. [MAT 2004 1B] Find the smallest value of
$f(x) = 2x^3 - 9x^2 + 12x + 3$
in the range $0 \le x \le 2$.

2.
$f\prime(x) = 6x^2 - 18x + 12 = 0$.
Divide by 6: $x^2 - 3x + 2 = 0$.
$(x-1)(x-2) = 0$, so $x = 1, 2$.
$f(1) = 8$ and $f(2) = 7$.
At the start of the interval, $f(0) = 3$.
Compare all stationary points and endpoints: the minimum value is 3, at $x = 0$.



Solutions to Extension Questions

3. [MAT 2001 1E] The maximum gradient of $y = x^4 - 4x^3 + 4x^2 + 2$ in $0 \le x \le 2\frac{1}{5}$ occurs when:
A) $x = 0$ B) $x = 1 - \frac{1}{\sqrt{3}}$ C) $x = 1 + \frac{1}{\sqrt{3}}$ D) $x = 2\frac{1}{5}$.

3.
Gradient: $\frac{dy}{dx} = 4x^3 - 12x^2 + 8x$. We must maximise this cubic.
$\frac{d^2y}{dx^2} = 12x^2 - 24x + 8 = 0 \Rightarrow 3x^2 - 6x + 2 = 0$.
Hence $x = 1 \pm \frac{1}{\sqrt{3}}$.
The gradient has a local maximum at $x = 1 - \frac{1}{\sqrt{3}}$, and a local minimum at $x = 1 + \frac{1}{\sqrt{3}}$.
Compare the stationary values and both interval endpoints:
At $x = 0$, $\frac{dy}{dx} = 0$.
At $x = 1 - \frac{1}{\sqrt{3}}$, $\frac{dy}{dx} = \frac{8}{3\sqrt{3}} \approx 1.540$.
At $x = 1 + \frac{1}{\sqrt{3}}$, $\frac{dy}{dx} = -\frac{8}{3\sqrt{3}}$.
At $x = \frac{11}{5}$, $\frac{dy}{dx} = \frac{264}{125} = 2.112$.
The maximum gradient therefore occurs at $x = 2\frac{1}{5}$. Answer: D.

4. [STEP I 2007 Q8] A curve is given by:
$y = ax^3 - 6ax^2 + (12a + 12)x - (8a + 16)$
where a is a real number. Show that this curve touches the curve with equation $y = x^3$ at $(2, 8)$. Determine the coordinates of any other point of intersection of the two curves.
(i) Sketch on the same axes these two curves when $a = 2$.
(ii) ... when $a = 1$
(iii) ... when $a = -2$

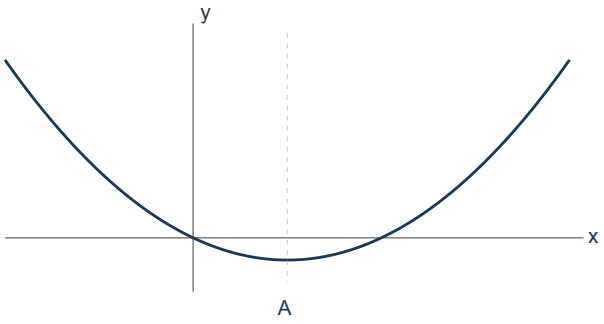
4
Consider $f(x) = ax^3 - 6ax^2 + (12a + 12)x - (8a + 16)$.
Since $f(2) = 8$ and $f'(2) = 12$, the curve $y = f(x)$ touches $y = x^3$ at $(2, 8)$: notice that both calculations are necessary to prove that the curves touch.
To find the other intersection point, let $f(x) = x^3$.
$\Rightarrow (a - 1)x^3 - 6ax^2 + (12a + 12)x - (8a + 16) = 0$.
Substituting $x = 2$: $8(a - 1) - 24a + 2(12a + 12) - (8a + 16) = 0$.
$\Rightarrow (x - 2)[(a - 1)x^2 - (4a + 2)x + (4a + 8)] = 0$ (notice that factorising by inspection is much easier than using a method such as long division).
Substituting $x = 2$ into the quadratic factor:
$4(a - 1) - 2(4a + 2) + (4a + 8) = 0$.
$\Rightarrow (x - 2)(x - 2)[(a - 1)x - (2a + 4)] = 0$.
So the other intersection point has coordinates $\left(\frac{2a+4}{a-1}, \left[\frac{2a+4}{a-1}\right]^3\right)$.
(i) When $a = 2$, $\frac{2a+4}{a-1} = 8$.
Hence the two graphs touch at $(2, 8)$ and intersect at $(8, 512)$.
$y = 2x^3 - 12x^2 + 36x - 32$ has no turning points: consider the derivative $6x^2 - 24x + 36 = 0$.
(ii) When $a = 1$, $\frac{2a+4}{a-1}$ is undefined.
Hence the two graphs touch at $(2, 8)$ and do not intersect elsewhere.
$y = x^3 - 6x^2 + 24x - 24$ has no turning points: consider the derivative $3x^2 - 12x + 24 = 0$.
(iii) When $a = -2$, $\frac{2a+4}{a-1} = 0$.
Hence the two graphs touch at $(2, 8)$ and intersect at $(0, 0)$.
$y = -2x^3 + 12x^2 - 12x$ turns when $x = 2 \pm \sqrt{2} \approx 3.4$ and 0.6: consider the derivative $-6x^2 + 24x - 12 = 0$.

Printed curve sketches: appendix pages 40·42.

Sketching Gradient Functions

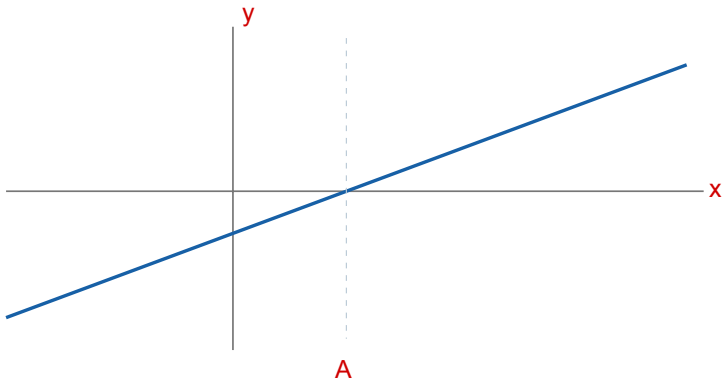
Solution 64

Original curve: $y = f(x)$



Gradient function: $y = f\prime(x)$

Gradient graph



The curve is a parabola with its minimum at A. Sketch its gradient function.

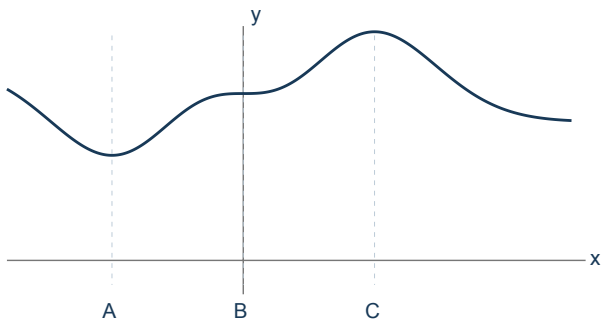
Before A the gradient is negative and increasing; at A it is zero; after A it is positive and increasing.

The derivative of a quadratic is a straight line. Its x-intercept is directly below the turning point.

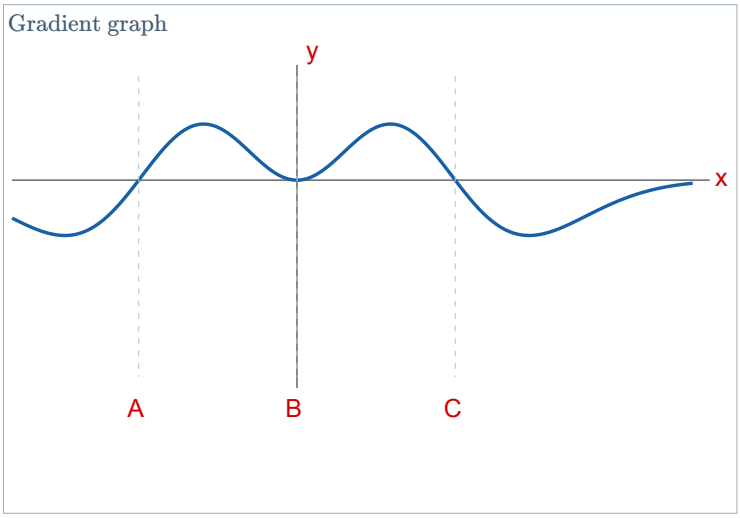
A Harder One

Solution 65

Original curve: $y = f(x)$



Gradient function: $y = f\prime(x)$



Sketch the gradient function. A is a local minimum, B a stationary inflection and C a local maximum.

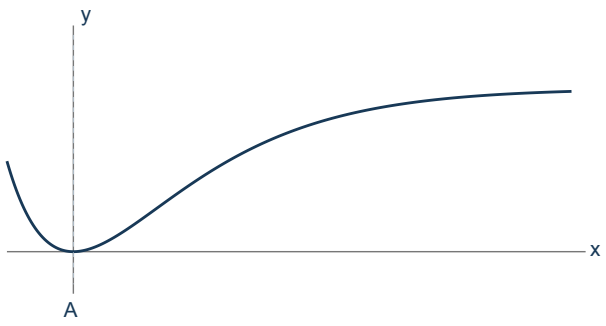
The gradient is negative before A, positive from A to C except at B, and negative after C. It tends to zero on the far right.

The derivative crosses the x-axis at A and C but touches it at B. Inflections of the original curve correspond to turning points of the gradient graph in this example.

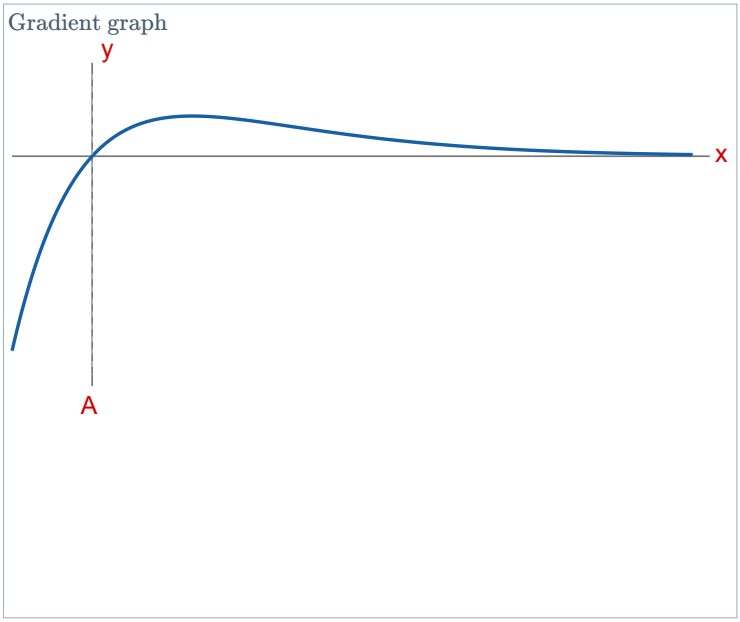
Test Your Understanding

Solution 66

Original curve: $y = f(x)$



Gradient function: $y = f\prime(x)$



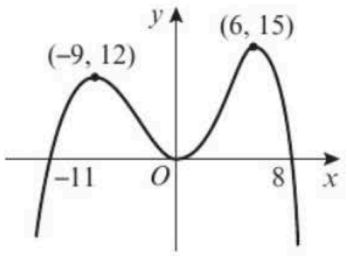
Sketch the gradient function of the curve shown. The curve has a minimum at A and approaches a horizontal asymptote to the right.

The gradient is negative to the left of A and zero at A. On the right it is positive, rises to a maximum and then tends to zero.

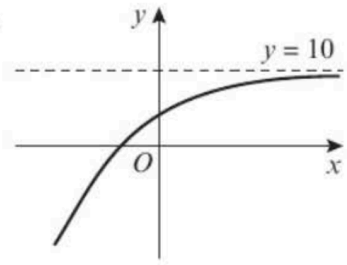
Exercise 12J – Pearson Pure Y1 page 278 — Solution 67

1 For each graph given, sketch the graph of the corresponding gradient function on a separate set of axes. Show the coordinates of any points where the curve cuts or meets the x-axis, and give the equations of any asymptotes.

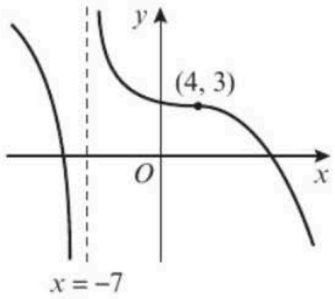
a



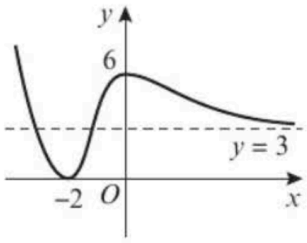
b



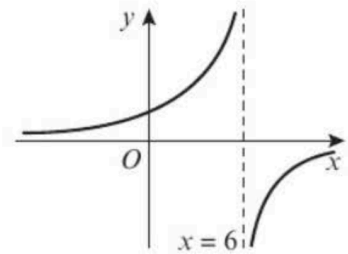
c



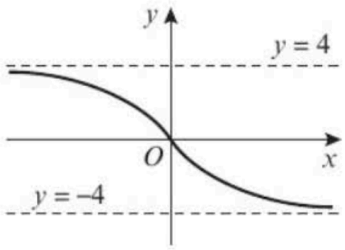
d



e



f



Printed worked solutions: appendix pages 43·46.

Exercise 12J – Pearson Pure Y1 page 278 — Solution 68

2. $f(x) = (x+1)(x-4)^2$.

a. Sketch the graph of $y = f(x)$.

b. On a separate set of axes, sketch the graph of $y = f\prime(x)$.

c. Show that $f\prime(x) = (x-4)(3x-2)$.

d. Use the derivative to determine the exact coordinates where the gradient function cuts the coordinate axes.

a.

$f(x) = (x+1)(x-4)^2 = x^3 - 7x^2 + 8x + 16$.

$y = 0 \Rightarrow x = -1$ or 4.

Stationary points: $\frac{dy}{dx} = 0$.

$\frac{dy}{dx} = 3x^2 - 14x + 8$.

$(3x-2)(x-4) = 0 \Rightarrow x = \frac{2}{3}$ or 4.

$x = \frac{2}{3}$: $y = (\frac{2}{3}+1)(\frac{2}{3}-4)^2 = \frac{500}{27}$.

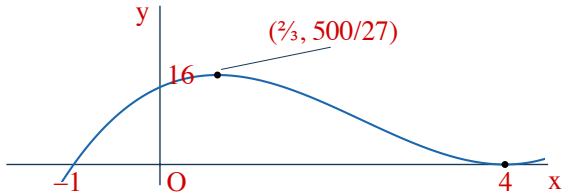
$x = 4$: $y = (4+1)(4-4)^2 = 0$.

Stationary points: $(\frac{2}{3}, \frac{500}{27})$ and $(4, 0)$.

$\frac{d^2y}{dx^2} = 6x - 14$.

$x = \frac{2}{3}$: $6(\frac{2}{3}) - 14 = -10 < 0$, so a maximum.

$x = 4$: $6(4) - 14 = 10 > 0$, so a minimum.



b.

x	$f(x)$	$f\prime(x)$
$x < \frac{2}{3}$	Positive gradient	Above x-axis
$x = \frac{2}{3}$	Maximum	Cuts x-axis
$\frac{2}{3} < x < 4$	Negative gradient	Below x-axis
$x = 4$	Minimum	Cuts x-axis
$x > 4$	Positive gradient	Above x-axis

c.

$f(x) = x^3 - 7x^2 + 8x + 16$.

$f\prime(x) = 3x^2 - 14x + 8$

$= (3x-2)(x-4)$.

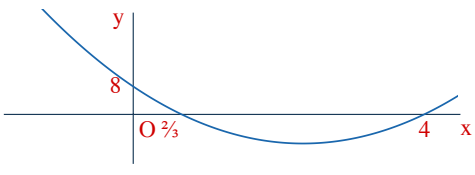
d.

$f\prime(x) = 3x^2 - 14x + 8 = 0$.

$(3x-2)(x-4) = 0 \Rightarrow x = \frac{2}{3}$ or 4.

When $x = 0$, $f\prime(0) = 8$.

Intercepts: $(\frac{2}{3}, 0)$, $(4, 0)$, $(0, 8)$.



Extension

Solution 69

[MAT 2015 1B] $f(x) = (x + a)^n$, where a is real and n is an integer with $n \geq 2$. How many intersections do $y = f(x)$ and $y = f\prime(x)$ have?

(a) Always an odd number. (b) Always an even number.

(c) Depends on a only. (d) Depends on n only. (e) Depends on both.

Answer

$(x + a)^n = n(x + a)^{n-1}$

$(x + a)^{n-1}(x + a - n) = 0$

$x = -a \quad \text{or} \quad x = n - a$

These are distinct because n $\geq$ 2. There are always two intersections: answer (b).

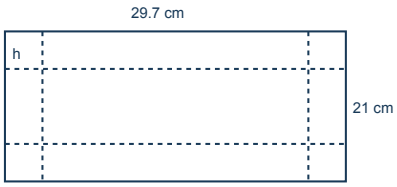
Factor the equation rather than dividing by a power of x + a, which would lose the intersection at x = −a.

Optimisation Problems/Modelling

Solution 70

Optimisation means choosing a variable to maximise or minimise a quantity.

1. An open cuboid from A4 paper



Cut squares of side h from the four corners of a 21 cm × 29.7 cm sheet, then fold up the sides. Which h maximises the volume?

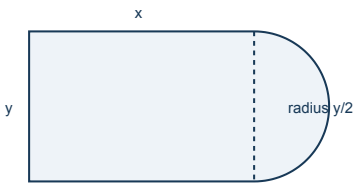
1

$V = h(21 - 2h)(29.7 - 2h)$

$V\prime = 623.7 - 202.8h + 12h^2$

$h \approx 4.04 \text{ cm} \quad (0 < h < 10.5)$

2. A pen with a semicircular end



A rectangular section x by y has a semicircle of diameter y at one end. The outer fence has length 50 m. Which x and y maximise the enclosed area?

2

$2x + y + \frac{\pi y}{2} = 50$

$A = xy + \frac{\pi y^2}{8} = 25y - \frac{4 + \pi}{8}y^2$

$y = \frac{100}{4 + \pi}, \qquad x = \frac{50}{4 + \pi}$

‘Rate of change’ Solution 71

A derivative is a rate of change

dy/dx means the rate at which y changes with respect to x. The same notation works for other physical quantities.

A sewage container fills at 20 cubic centimetres per second. Use appropriate notation to represent this.

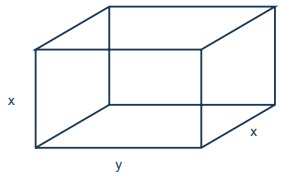
Rate

$$\frac{dV}{dt} = 20 \text{ cm}^3/\text{s}$$

Here V is volume and t is time. The derivative describes the rate at which the volume changes with time.

Example Optimisation Problem Solution 72

An open-topped cuboid tank is made from 54 m^2 of sheet metal. Its height is x metres and two opposite vertical faces are squares. Let the remaining dimension be y metres.



(a) Show that $V = 18x - \frac{2}{3}x^3$.

(b) Find and justify the maximum volume.

a

$$2x^2 + 3xy = 54, \qquad V = x^2 y$$

$$y = \frac{54 - 2x^2}{3x}$$

$$V = x^2 \frac{54 - 2x^2}{3x} = 18x - \frac{2}{3}x^3$$

b

$$\frac{dV}{dx} = 18 - 2x^2 = 0 \Rightarrow x = 3$$

$$\frac{d^2V}{dx^2} = -4x = -12 < 0$$

$$V = 18(3) - \frac{2}{3}(3)^3 = 36 \text{ m}^3$$

Test Your Understanding — Solution 73

A cuboid has a rectangular cross-section whose length is twice its width, x cm, as shown in Figure 2. The volume is 81 cm^3.

The total length of the twelve edges is L cm.

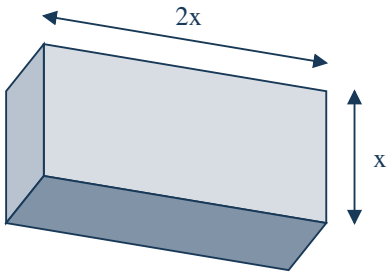


Figure 2

a. Show that $L = 12x + \frac{162}{x^2}$. (3 marks)

b. Use calculus to find the minimum value of L. (6 marks)

c. Justify, by further differentiation, that your value of L is a minimum. (2 marks)

a.

Let the remaining dimension be y cm.

Volume: $2x^2y = 81$, so $y = \frac{81}{2x^2}$.

Total edge length:

$L = 2(2x + x + 2x + x) + 4y$

$= 12x + 4y$.

Substitute for y:

$L = 12x + 4\left(\frac{81}{2x^2}\right)$

$= 12x + \frac{162}{x^2}$.

b.

$\frac{dL}{dx} = 12 - \frac{324}{x^3} = 12 - 324x^{-3}$.

At a stationary value, $\frac{dL}{dx} = 0$.

$12 - \frac{324}{x^3} = 0$.

$x^3 = \frac{324}{12} = 27$.

Since $x > 0$, $x = \sqrt[3]{27} = 3$.

$L = 12(3) + \frac{162}{3^2} = 54$ cm.

c.

$\frac{d^2L}{dx^2} = \frac{972}{x^4}$.

At $x = 3$,

$\frac{d^2L}{dx^2} = \frac{972}{3^4} = 12 > 0$.

Therefore $L = 54$ cm is a minimum.

Exercise 12K – Pearson Pure Y1 page 281 — Solution 74

6. A rectangular garden is fenced on three sides; the house forms the fourth side. The fence is 80 m long. Let y be the distance from the house and x the width.



(a) Show that A = y(80 − 2y).

(b) Find the dimensions and maximum area.

a

$x + 2y = 80 \Rightarrow x = 80 - 2y$

$A = xy = y(80 - 2y)$

b

$\frac{dA}{dy} = 80 - 4y = 0 \Rightarrow y = 20$

$x = 40, \quad \frac{d^2A}{dy^2} = -4 < 0$

$A_{max} = 40 \times 20 = 800$ m^2

Exercise 12K – Pearson Pure Y1 page 281 Solution 75

7. A closed cylinder has total surface area 600π cm^2. Its radius is r cm and its height is h cm.

(a) Show that $V = 300\pi r - \pi r^3$.

(b) Find the maximum volume.

a

$$2\pi rh + 2\pi r^2 = 600\pi$$

$$rh = 300 - r^2$$

$$V = \pi r(rh) = 300\pi r - \pi r^3$$

b

$$\frac{dV}{dr} = 300\pi - 3\pi r^2 = 0$$

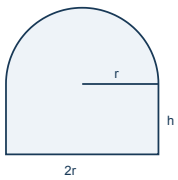
$$r = 10, \quad h = 20$$

$$\frac{d^2V}{dr^2} = -6\pi r < 0$$

$$V_{\max} = 2000\pi \text{ cm}^3$$

Exercise 12K – Pearson Pure Y1 page 281 Solution 76

9. A shape consists of a rectangle with a semicircular top. Its perimeter is 40 cm; the semicircle has radius r cm.



(a) Show that $A = 40r - 2r^2 - \frac{1}{2}\pi r^2$. (2 marks)

(b) Hence find the maximum area. (4 marks)

a

$$2h + 2r + \pi r = 40$$

$$h = 20 - r - \frac{\pi r}{2}$$

$$A = 2rh + \frac{\pi r^2}{2} = 40r - 2r^2 - \frac{\pi r^2}{2}$$

b

$$\frac{dA}{dr} = 40 - (4 + \pi)r = 0$$

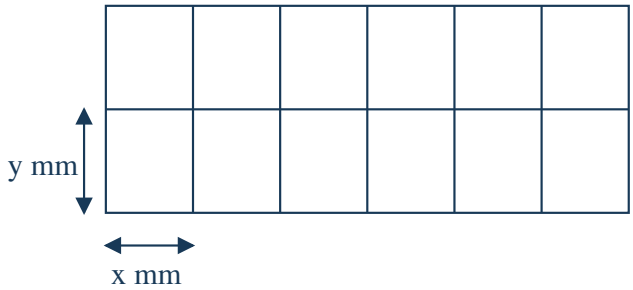
$$r = \frac{40}{4 + \pi}, \quad \frac{d^2A}{dr^2} = -(4 + \pi) < 0$$

$$A_{\max} = \frac{800}{4 + \pi} \text{ cm}^2$$

Exercise 12K – Pearson Pure Y1 page 281

Solution 77

10. A wire frame forms a large rectangle divided into 12 equal rectangles, in six columns and two rows. Each small rectangle has width x mm and height y mm. The total length of wire is 1512 mm.



a. Show that the total area is $A = 1296x - \frac{108x^2}{7}$ mm^2. (4 marks)

b. Find the maximum area enclosed by the frame. (4 marks)

a.

Three horizontal wires each have length $6x$.

Seven vertical wires each have length $2y$.

Total wire length $= 18x + 14y$ mm.

$18x + 14y = 1512$.

$y = \frac{1512-18x}{14}$.

Total area: $A = 2y \times 6x = 12xy$.

Substitute for y:

$A = 12x\left(\frac{1512-18x}{14}\right)$

$= 1296x - \frac{108x^2}{7}$.

b.

For a maximum, $\frac{dA}{dx} = 0$.

$\frac{dA}{dx} = 1296 - \frac{216}{7}x$.

$1296 - \frac{216}{7}x = 0$.

$x = \frac{7\times1296}{216} = 42$.

When $x = 42$,

$A = 1296 \times 42 - \frac{108}{7} \times 42^2$

$= 27\,216$.

Maximum area $= 27\,216$ mm^2.

Check: $\frac{d^2A}{dx^2} = -\frac{216}{7} < 0$, so a maximum.

Extension

Solution 78

[STEP I 2006 Q2] A goat is tethered at ground level to a side of a square barn of side 2a in a large field. Its rope has length 4a. If A is the grazing area, prove that $A \leq 14\pi a^2$ and find its minimum.

Let x be the distance from one corner, with $0 \leq x \leq 2a$. The accessible area is a semicircle in front of the barn and four quarter-circle sectors reached around its corners.

$$A = \frac{\pi}{2}(4a)^2 + \frac{\pi}{4}\left[(4a-x)^2 + (2a+x)^2 + (2a-x)^2 + x^2\right]$$

$$= \pi[13a^2 + (x-a)^2]$$

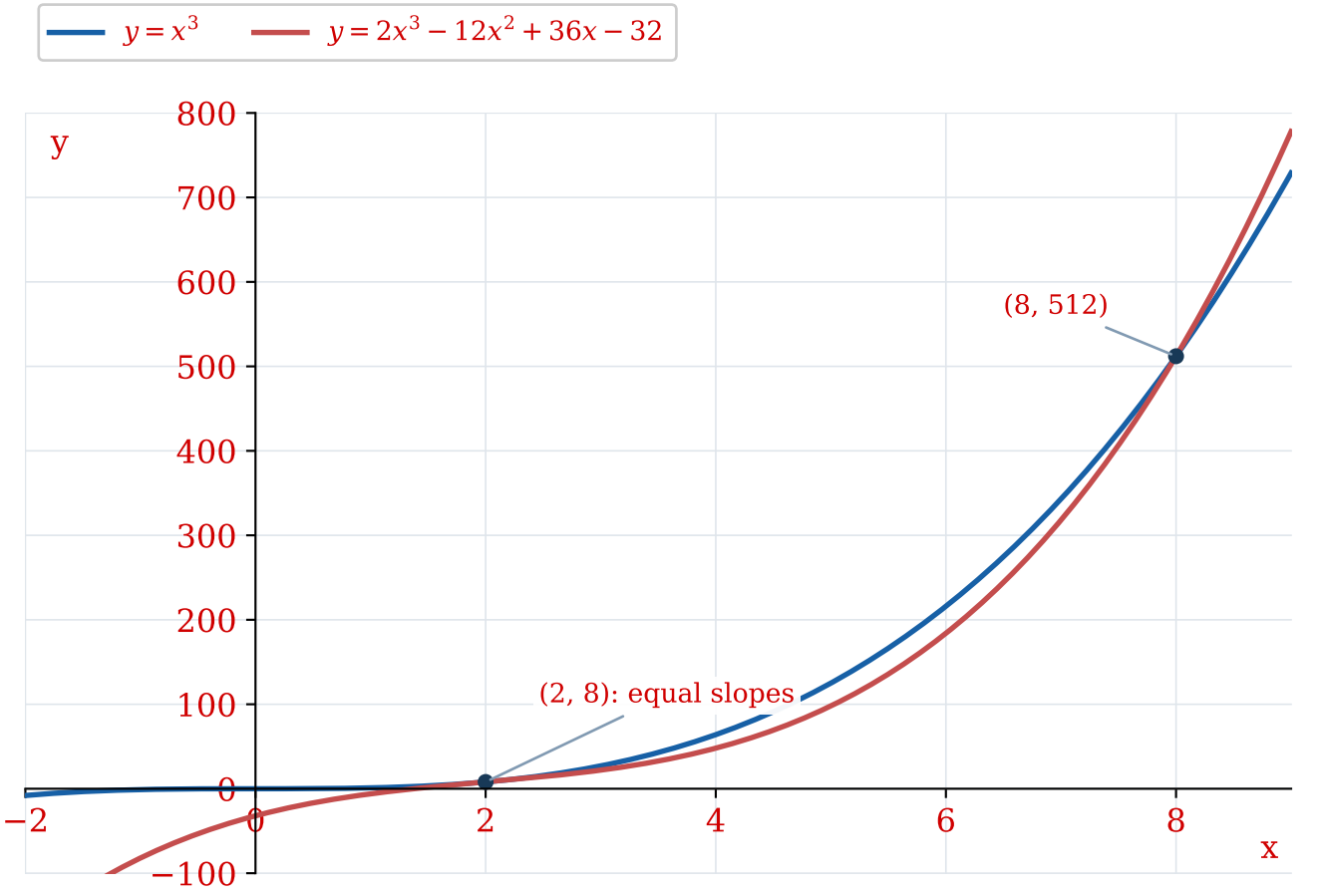
Find the bounds

$0 \leq (x-a)^2 \leq a^2$

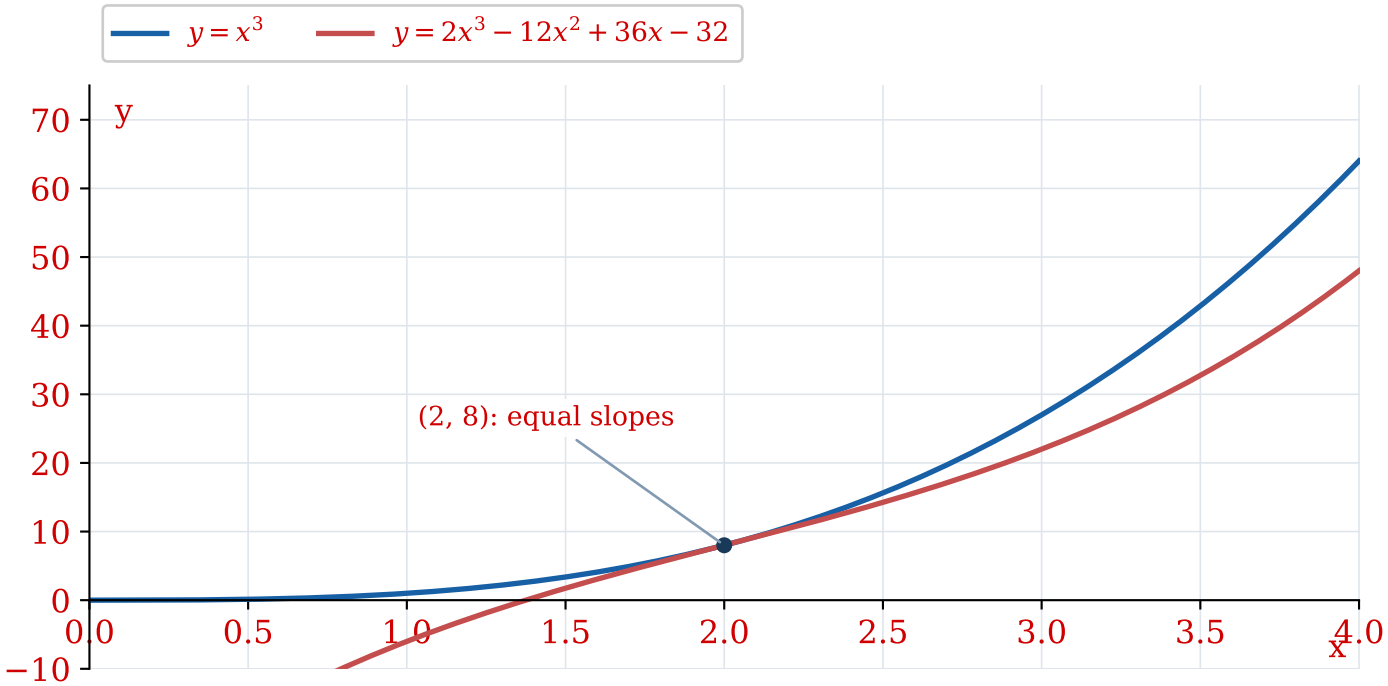
$13\pi a^2 \leq A \leq 14\pi a^2$

The minimum occurs at the middle of a side (x = a). The maximum occurs at a corner (x = 0 or 2a).

Case (i): a = 2



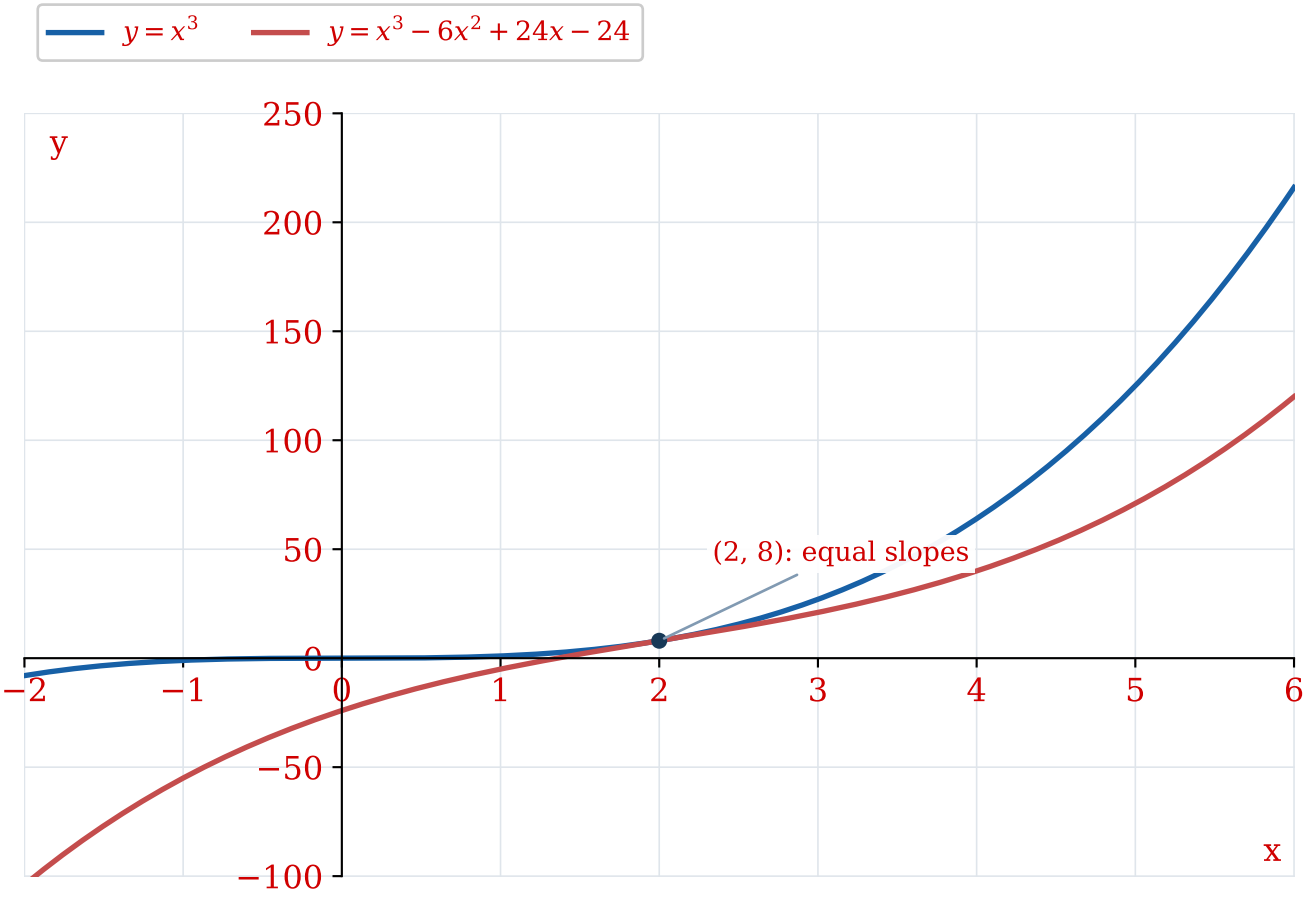
Detail around the touching point



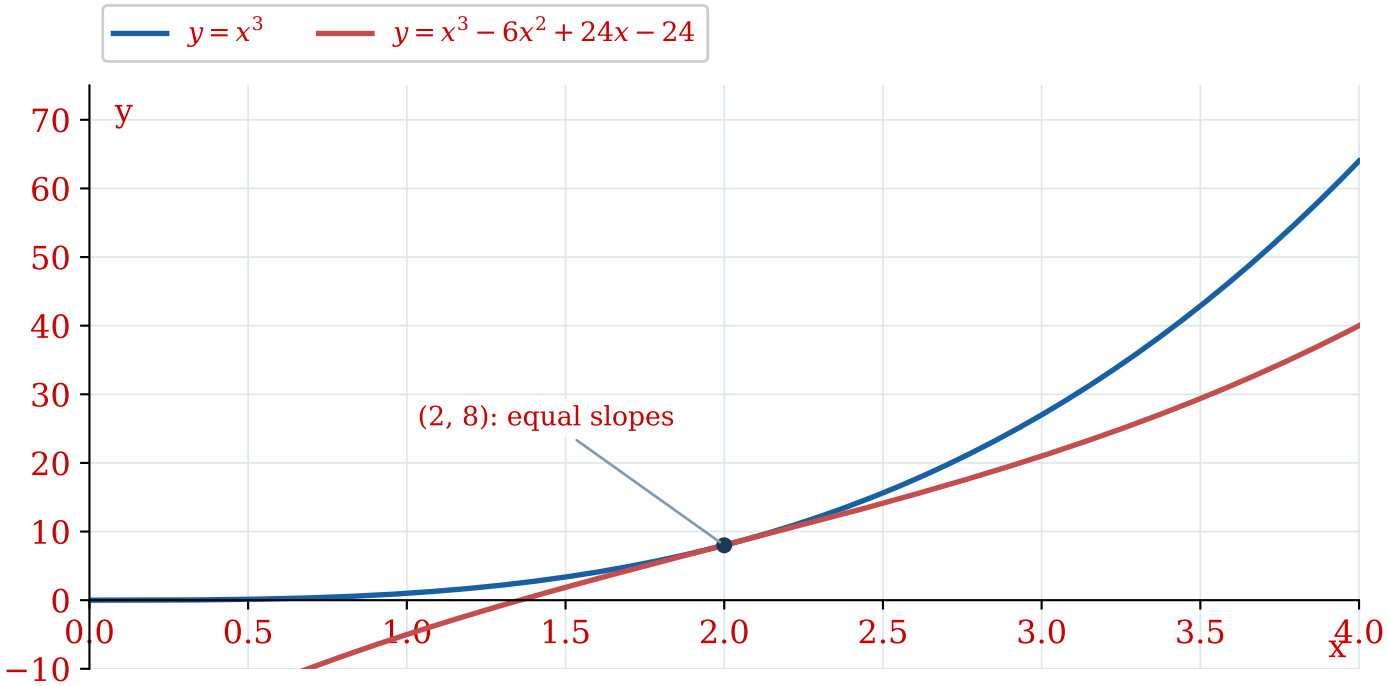
The curves touch at (2, 8) and also intersect at (8, 512).
The red curve has derivative $6(x - 2)^2 + 12 > 0$, so it has no turning points.

FMQ · Calculated from the supplied original question · Axis scales are labelled.

Case (ii): a = 1



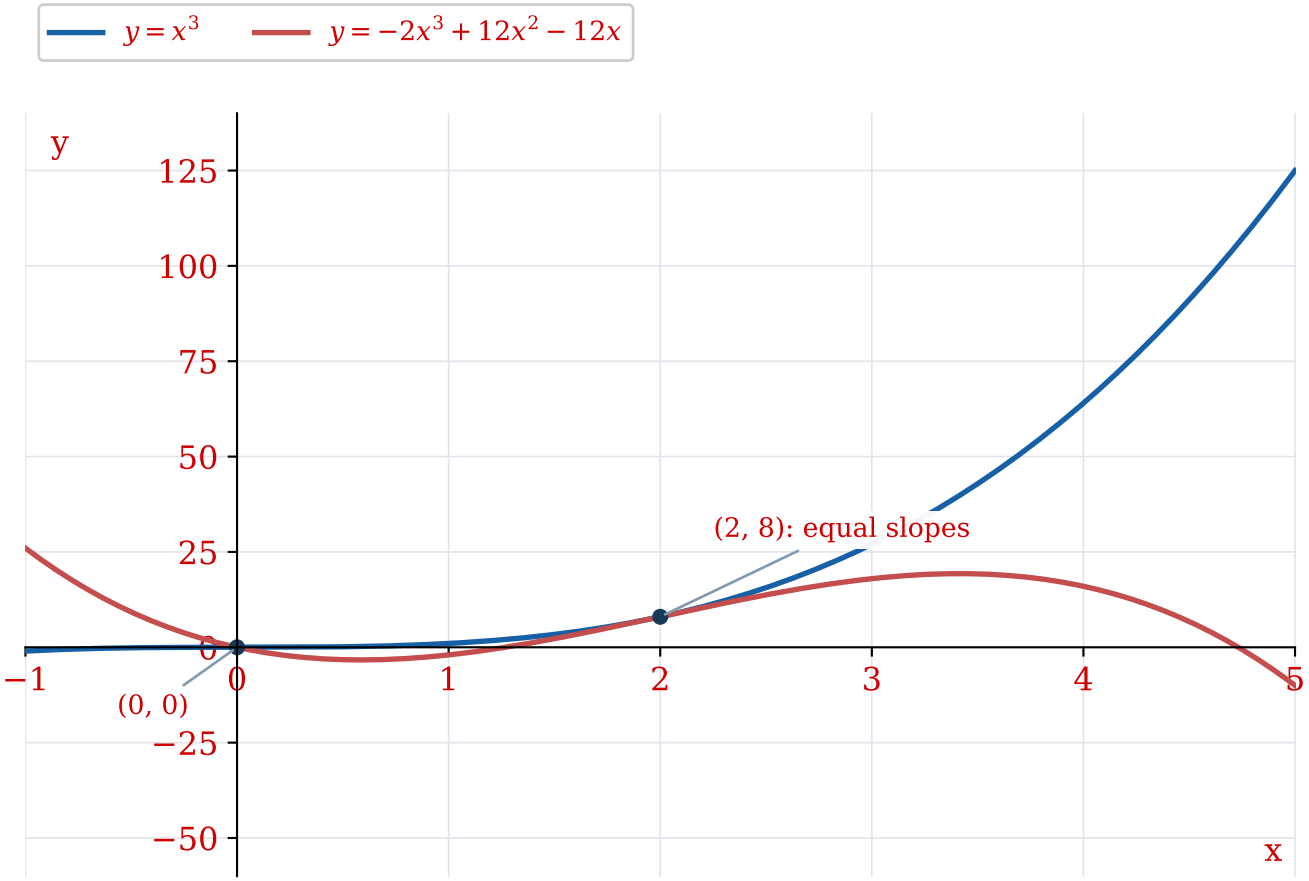
Detail around the touching point



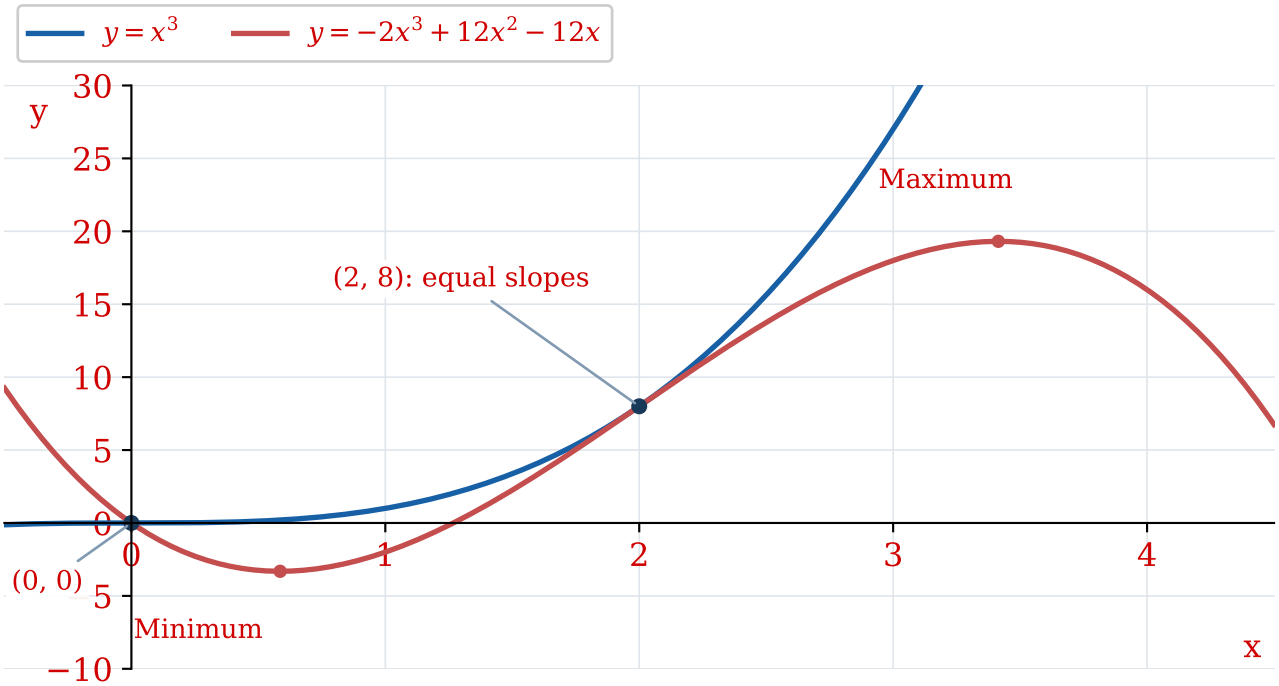
The curves touch at (2, 8) and have no other intersection.
The red curve has derivative $3(x - 2)^2 + 12 > 0$, so it has no turning points.

FMQ · Calculated from the supplied original question · Axis scales are labelled.

Case (iii): a = -2



Detail showing the two turning points

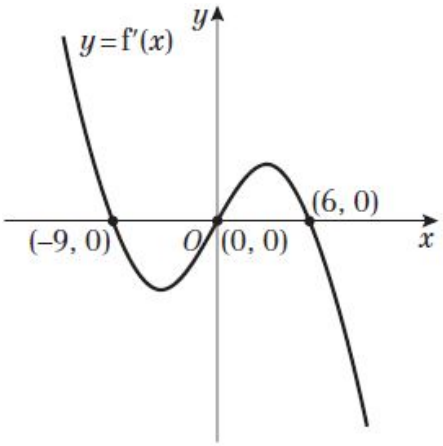


The curves touch at (2, 8) and also intersect at (0, 0).
The red curve turns at x = 2 − √2 (minimum) and x = 2 + √2 (maximum).

Differentiation 12J

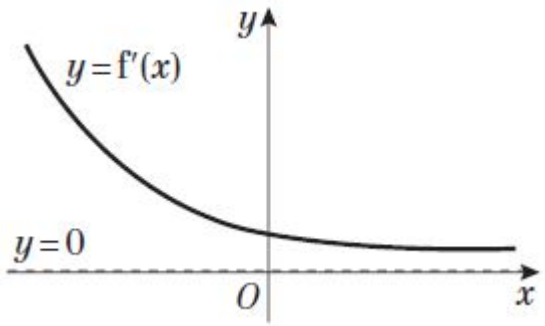
1 a

x	$y = f(x)$	$y = f'(x)$
$x < -9$	Positive gradient	Above x-axis
$x = -9$	Maximum	Cuts x-axis
$-9 < x < 0$	Negative gradient	Below x-axis
$x = 0$	Minimum	Cuts x-axis
$0 < x < 6$	Positive gradient	Above x-axis
$x = 6$	Maximum	Cuts x-axis
$x > 6$	Negative gradient	Below x-axis



b

x	$y = f(x)$	$y = f'(x)$
All values of x	Positive gradient	Above x-axis with asymptote at $y = 0$

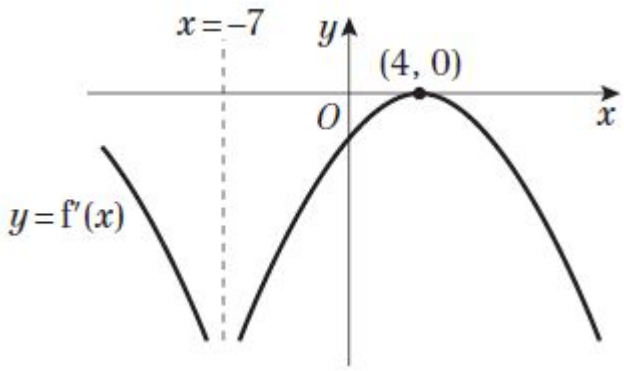


c

x	$y = f(x)$	$y = f'(x)$
$x < -7$	Negative gradient	Below x-axis with asymptote at $x = -7$
$-7 < x < 4$	Negative gradient	Below x-axis
$x = 4$	Point of inflection	Touches x-axis
$x > 4$	Negative gradient	Below x-axis

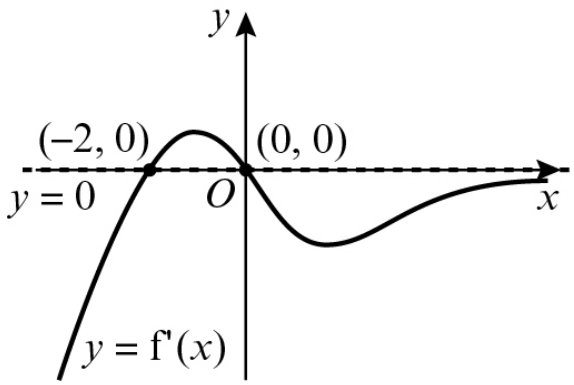
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1 c



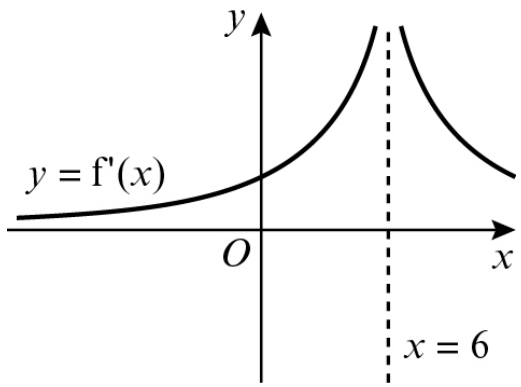
d

x	$y = f(x)$	$y = f'(x)$
$x < -2$	Negative gradient	Below x-axis
$x = -2$	Minimum	Cuts x-axis
$-2 < x < 0$	Positive gradient	Above x-axis
$x = 0$	Maximum	Cuts x-axis
$x > 0$	Negative gradient	Below x-axis with asymptote at $y = 0$



e

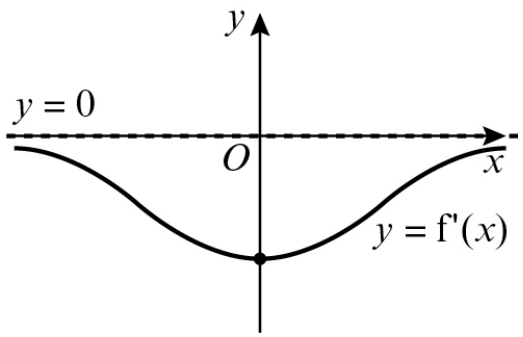
x	$y = f(x)$	$y = f'(x)$
$x < 6$	Positive gradient	Above x-axis with asymptote at $x = 6$
$x > 6$	Positive gradient	Above x-axis with asymptote at $x = 6$



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1 f

x	$y = f(x)$	$y = f'(x)$
$x < 0$	Negative gradient	Below x-axis with asymptote at $y = 0$
$x > 0$	Negative gradient	Below x-axis with asymptote at $y = 0$



2 a $y = f(x) = (x + 1)(x - 4)^2 = x^3 - 7x^2 + 8x + 16$

When $y = 0$, $x = -1$ or $x = 4$

To find stationary points, $\frac{dy}{dx} = 0$:

$\frac{dy}{dx} = 3x^2 - 14x + 8$

$(3x - 2)(x - 4) = 0$

$x = \frac{2}{3}$ or $x = 4$

When $x = \frac{2}{3}$, $y = \left(\frac{2}{3}+1\right)\left(\frac{2}{3}-4\right)^2 = \frac{500}{27}$

When $x = 4$, $y = (4 + 1)(4 - 4)^2 = 0$

So $\left(\frac{2}{3}, \frac{500}{27}\right)$ and $(4, 0)$ are stationary points.

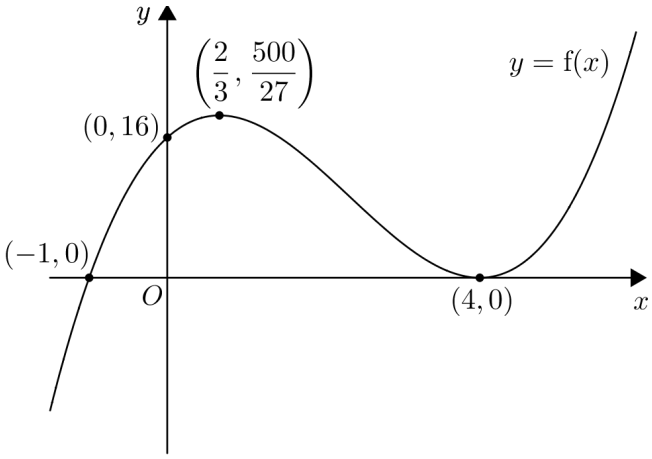
$\frac{d^2y}{dx^2} = 6x - 14$

When $x = \frac{2}{3}$, $\frac{d^2y}{dx^2} = 6\left(\frac{2}{3}\right) - 14 = -10 < 0$

So $\left(\frac{2}{3}, \frac{500}{27}\right)$ is a maximum.

When $x = 4$, $\frac{d^2y}{dx^2} = 6(4) - 14 = 10 > 0$

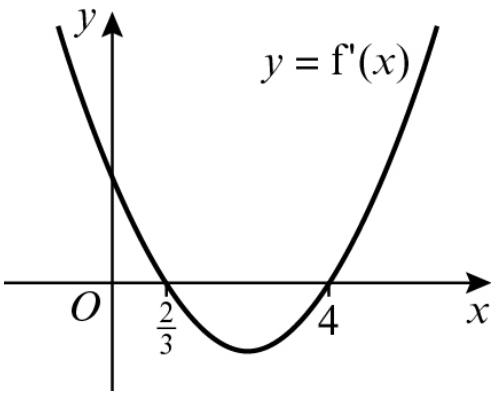
So $(4, 0)$ is a minimum.



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2 b

x	$y = f(x)$	$y = f'(x)$
$x < \frac{2}{3}$	Positive gradient	Above x-axis
$x = \frac{2}{3}$	Maximum	Cuts x-axis
$\frac{2}{3} < x < 4$	Negative gradient	Below x-axis
$x = 4$	Minimum	Cuts x-axis
$x > 4$	Positive gradient	Above x-axis



c $f(x) = (x + 1)(x - 4)^2 = x^3 - 7x^2 + 8x + 16$

$f'(x) = 3x^2 - 14x + 8$

$= (3x - 2)(x - 4)$

d $f'(x) = 3x^2 - 14x + 8$

$(3x - 2)(x - 4) = 0$

$x = \frac{2}{3}$ or $x = 4$

When $x = 0$, $f'(x) = 8$

The points where the gradient function cuts the axes are $\left(\frac{2}{3}, 0\right)$, (4, 0) and (0, 8).

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